

CHAPTER

7

PROVEN METHOD FOR SPECIFYING BOTH SPECTRAL ALARM BANDS AS WELL AS NARROWBAND ALARM ENVELOPES USING TODAY'S CONDITION MONITORING SOFTWARE SYSTEMS

By: James E. Berry, P.E.
Vice-President
Technical Associates of Charlotte, P.C.
Charlotte, NC

James E. Berry has 25 years mechanical engineering experience including 23 years specializing in Machinery Vibration Diagnostics, setup and implementation of Condition Monitoring Programs, Modal Analysis and Stress Analysis. Mr. Berry received both his Bachelor of Science (1973) and Masters Degrees (1974) in Mechanical Engineering from North Carolina State University. He is also a registered professional engineer and is an active member of the Vibration Institute. He has published several vibration analysis articles in technical journals such as *Sound and Vibration Magazine*, given technical papers and seminars to the Vibration Institute, and has given presentations to several engineering societies including ASME, AIPE, ASA, and ASTM. Serving as a Consulting Engineer, he has performed vibration analysis on a wide variety of both process and utility machinery for a diverse group of clientele served by Technical Associates. He has also authored four seminar texts focusing on vibration analysis and condition monitoring and has been providing professional training services for 12 years.

7.0 ABSTRACT

The fourth edition of this paper has been written primarily with the objective of not only expanding the coverage of Spectral Band Alarm setups to encompass additional machine types, but also to refine those previously established for machine types covered before. Comprehensive statistical analyses have been conducted to help the user specify meaningful overall alarm levels as well as spectral band alarm levels. Here again, this paper is intended to give the analyst a good starting point; such spectral bands and alarm levels should be reviewed after a sufficient quantity of surveys have been conducted, modifying those which are found to need refinement.

Although there is much literature available today on how to diagnose machine problems using vibration analysis, there is little material available on how to specify effective spectral alarm bands on various types of machinery. These spectral alarm bands are now offered within the software of several vendors serving the field of condition monitoring, and thanks to these vendors, provide the potential of detecting numerous machine problems that might otherwise go unnoticed. In the detection and analysis process, if these spectral bands are utilized, they can save the user thousands of dollars in maintenance expenditures and make significant impact on improving plant profitability. First, one needs to know that his machine has a problem. Then, he must take steps to diagnose both the source of the problem and determine its severity. The purpose of this paper is to provide a documented technique on how to specify peak velocity spectral alarm levels and frequency bands for measurements taken on the housings of general process and utility machinery. If properly specified for the specific machine type, drive configuration, bearing type and operating speed, these spectral

alarm bands will notify the user that he has a problem without generating a series of false alarm "emergencies". The techniques included are not intended, and will not apply to all machine types under all operating conditions. However, they have been successfully applied to a diverse array of machinery ranging from common pumps and blowers to refrigeration chillers, hammer mills, machine tool spindles, high-speed centrifugal air compressors, moderate speed rolling mill drives, etc.

This paper now applies not only to those software systems which allow a number of spectral alarm bands ("power bands"), but also to those which allow generation of narrowband spectral alarms which place a threshold envelope around individual frequencies which are caused by specific sources within the machine ("threshold band alarms").

The complete tabulated procedure for properly specifying spectral alarm bands is included in Table III. It is then followed by several examples which include complete specification of spectral alarm bands for various machine types.

7.1 INTRODUCTION TO SPECIFYING SPECTRAL ALARM BANDS & FREQUENCY RANGES

Properly specified spectral alarm bands are probably one of the most critical weapons in the condition monitoring arsenal today for detecting potentially serious problems which develop in machinery. However, although thousands of data collectors and software are now in place throughout the world having the capability of comparing each new FFT spectrum captured to these user-defined spectral alarm bands, surveys have shown that very few users have sufficient experience to know how to properly and effectively set up these bands in their computers. In fact, a large percentage of plants do not use spectral alarm band capabilities even though they are offered in their software. Instead, they depend on trending of overall levels to warn of impending machine problems. On the other hand, many of those who themselves have made concerted efforts to specify and use these bands often complain that they do not feel very comfortable with the bands they have specified; and do not have the time required to learn how to specify one set for one machine type and an entirely different set for another, depending on how the machine is configured (bearing type, operating speed, drive configuration, etc.). Many users at these plants are hard pressed just to determine what overall alarm vibration levels should be specified for these machines, much less have the time to research how to specify individual band frequency ranges and alarm levels. Many spend hundreds of man-hours just trying to specify the optimum overall, and even then, doubt how meaningful these are.

As we now know from spectrum analysis, if it were possible to obtain so-called "perfect overall vibration specifications", potentially serious problems can still develop within machines, and yet, can have negligible effect on the level of overall vibration. However, these same problems would be noticeable in the FFT spectrum. But, if no spectral alarm system were in place to even detect the presence of a problem, the user would be unaware of its existence until possibly considerable damage had been done, not only to one component, but possibly to several other components in this machine as well.

Such problems as deteriorating bearings, gears, electrical problems, etc. may not make themselves known for some time to those who depend only on overall levels to detect problems. For example, a bearing defect frequency might grow by a factor of 4X from .03 to .12 in/sec and cause almost no change in the overall if amplitudes at 1X and 2X RPM were, say .35 in/sec and .20 in/sec (in this case, the 4X increase by the bearing frequency would only increase the overall from approximately .40 to .42 in/sec). Even though there would be a definite increase in bearing defect vibration, the overall alone would simply not be sufficiently sensitive to show there was any real change.

Therefore, the express purpose of this paper is to offer a procedure for specifying meaningful spectral alarm bands for a variety of machine types and configurations. This method has been in use by our company for numerous years as we have set up and carried out complete condition monitoring programs for a large number of clients in a diverse array of industries. This technique is summarized in Table III.

This technique is not meant (or implied) to specify concrete spectral alarm bands that cannot be altered, no matter what the unique spectral characteristics of a given machine family. Instead, it is meant to provide the analyst with a firm starting point to:

- (1) Allow a plant having no prior experience or machine vibration history to initially set up effective spectral alarm bands for hundreds of machines in his plant prior to making baseline (or initial) measurements;
- (2) Allow a more experienced plant to set up spectral alarm bands for the first time, even though the plant might have captured data on large numbers of machines for several years, but has never set up the bands due to a lack of understanding on how to properly do so;
- (3) Allow the plant which in fact has installed spectral alarm bands to objectively compare them to other setups, and to evaluate how effective their current bands are.

Please keep in mind that after several samples of data have been acquired, the user should carefully review how each setup for each grouping of machines is working (assuming it is possible to place all his machines into specific groupings or families). Included in the paper will not only be how to specify spectral alarm bands, but also suggestions on how to evaluate their effectiveness and refine them as well.

7.11 TWO TYPES OF SPECTRAL ALARM BANDS

Importantly, note that two different types of spectral alarm bands are used by several different condition monitoring software versions - 1) Absolute Threshold, and 2) so-called "Power Band" type. It is important that the user know which type is employed by his predictive maintenance software system and take this into account.

Absolute threshold systems enable users to specify the maximum allowable amplitude of any single peak within each band. If any peak equals or exceeds this threshold, it will cause the band to go into alarm.

On the other hand, power band systems calculate the total energy (or "power") within each band generated by all the peaks within that band. The total power within each band is calculated using the same equation as that used to determine the overall level of an entire spectrum as per the following formula:

$$OA = \frac{\sqrt{\sum_{i=1}^n A_i^2}}{\sqrt{N_{BF}}} = \frac{\sqrt{A_1^2 + A_2^2 + A_3^2 + \dots + A_N^2}}{\sqrt{1.5}} \quad \text{EQUATION (1)}$$

Where:

OA = Overall Level of Spectrum or Band
 n = Number of FFT Lines of Resolution
 A_i = Amplitude of Each of the FFT Lines
 N_{BF} = Noise Bandwidth for Window Chosen
 1.5 for Hanning Window

$$\text{Which means } OA = .8165 \sqrt{\sum_{i=1}^n A_i^2}$$

Note that it is not necessary for any individual peak to equal or exceed a power band alarm in order to exceed the alarm for that band. That is, if a band were specified to extend from .50X F_{MAX} through F_{MAX} using a 400 line FFT, Equation (1) would be used to calculate the power from the 200th line through the 400th line of the spectrum. For example, for a power band having a .20 in/sec specified alarm, the energy of only 2 peaks within this band having amplitudes of .175 and .185 in/sec, respectively, would likely exceed this alarm.

Importantly, the spectral alarm band amplitudes specified in Table III assume the power band type. The same frequency ranges would apply for either alarm band system. However, if one had absolute threshold bands, he should lower alarm levels specified in Table III somewhat - probably on the order of 20% to 30%, particularly for those bands having wider frequency ranges.

7.12 WHICH VIBRATION PARAMETER TO USE IN SPECTRAL ALARM BANDS - DISPLACEMENT, VELOCITY OR ACCELERATION?

Three important items must be understood when setting up spectral alarm bands for machines. First, one must know what forcing frequencies will be generated by such things as rolling element bearing wear, sleeve bearing wear, gear problems, electrical problems, unbalance, misalignment, etc. Secondly, however, he must then know which vibration parameter (displacement, velocity or acceleration) will best detect those problems he will see on his particular machines. Thirdly, he must know how many FFT lines must be used to even show the presence of such problems.

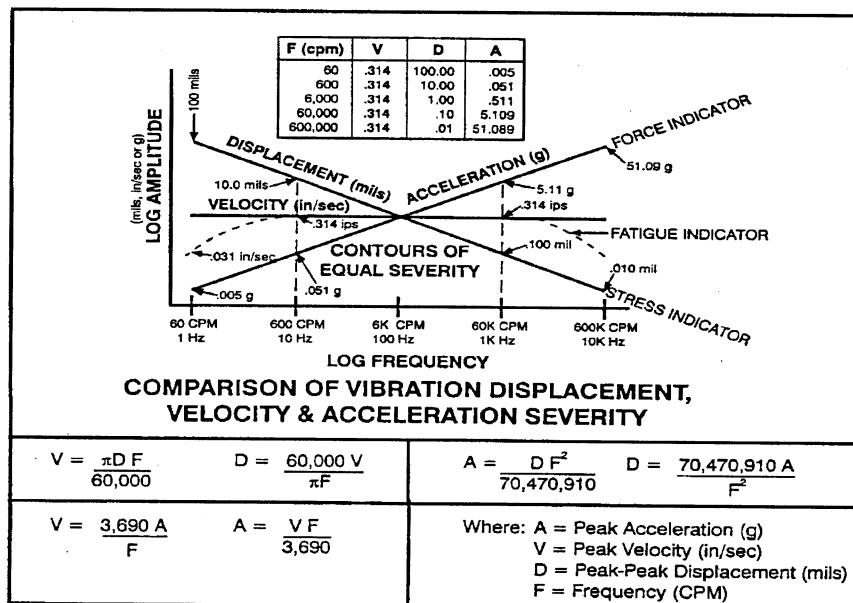


FIGURE 1
CONTOURS OF EQUAL SEVERITY AND CONVERSION FORMULAS

For example, some users still acquire displacement spectra for most all of their machinery because "this is the way it has been done for many years". Figure 1 shows that while displacement does a good job on (and is the most sensitive parameter to) low frequency measurements predominantly below 600 CPM, it does not adequately detect problems which are higher frequency in nature such as rolling element bearing and gear wear. For example, assume a belt-driven blower with a nominal 3600 RPM motor, having a serious bearing defect frequency amplitude of .30 in/sec (peak) at 60,000 CPM, as well as a .30 in/sec amplitude at running speed (1X RPM). If peak-peak displacement were used to evaluate this machine, the equations in Figure 1 show that the .30 in/sec level at 60,000 CPM would correspond to a deceptively low amplitude of only .095 mil at

60,000 CPM; while the running speed vibration would be approximately 1.59 mils at 3600 CPM. Since normal alarms for this machine would be set at approximately 2 mils, the 3600 RPM running speed vibration would readily be visible in a linear amplitude spectrum, while that at the 60,000 CPM bearing frequency would not. This would give one a false sense of security concerning his bearing health. He may not even notice the presence of the bearing defect peak, and therefore, would not even attempt to determine its source. On the other hand, velocity spectra would clearly display the bearing frequency peak and would show it at just as high in amplitude as that at operating speed in this case (but in reality, of much greater problem severity than that at 1X RPM).

For another example, even though displacement would be the best indicator of unbalance or misalignment on a machine running only 300 RPM, if one's primary interest were rolling element bearing condition, velocity would again be the best parameter to employ. This is commonly the case with paper machines and other large, low-speed machinery.

On the other hand, velocity spectra have their limitations as well. For example, assume a common centrifugal air compressor running at 3580 RPM and having a 344 tooth bullgear. This machine would have a fundamental gear mesh frequency (GMF) of approximately 1,231,500 CPM (20,525 Hz). Experience with these machines proves, that not only must one evaluate amplitude at the fundamental gear mesh frequency, but at least at the second and third GMF harmonic as well. A good conditioned, well aligned set of gears will normally have a level of approximately 6 g (peak) at 2X GMF at approximately 2,463,000 CPM (41,050 Hz). If these gears were to develop definite problems increasing the amplitude 10 times higher to 60 g at 2X GMF, this would correspond to a peak velocity of only .089 in/sec (and an even more deceptive peak-peak displacement of only .00070 mil).

Therefore, the best indicator of problems which generate forcing frequencies in these high frequency regions, particularly above 300,000 CPM, is acceleration.

However, velocity spectra will prove to be the best indicator of a large majority of problems likely on about 80% to 90% of rotating machinery. Therefore, the emphasis of this article will be on specification of peak velocity spectral alarm bands (again, one can easily convert peak velocity overall amplitudes shown in Table II and spectral alarm band amplitudes shown in Table III to RMS simply by multiplying levels by a factor of .707). In fact, the great majority of instruments available actually make RMS measurements, and "convert" them to so-called peak velocity (or peak acceleration) by simply multiplying the amplitude measurement for each frequency by 1.414 ($\sqrt{2}$).

Similar bands can be specified for either displacement or acceleration for those machines requiring these parameters. Then, it will be important to take into account how displacement and acceleration vary with the frequency (see formulas in Figure 1). For example, assume that when using velocity, one were going to set the alarm level for 1X RPM at .30 in/sec, while setting the alarms at 2X and 3X RPM at one half this amount, or .15 in/sec. Since displacement varies directly with velocity, but inversely with frequency (see Figure 1), if he sets the 1X RPM displacement alarm at 6 mils, he will need to set the 2X RPM alarm level at only 1/4 that at 1X RPM (1.5 mil) and the 3X RPM alarm at only 1/6 that at 1X RPM (1 mil).

On the other hand, given the same machine, if one wished to convert from velocity to acceleration bands, he would need to make the relationship of one harmonic to another in a manner opposite that for displacement. For example, if the velocity alarms for 2X and 3X RPM have been set at one half that at 1X RPM, he will need to set the acceleration alarm at 2X RPM at the same level as that at 1X RPM, but set the 3X alarm at 50% higher than that at 1X RPM. This is due to the fact that acceleration varies directly with both velocity and frequency (see formula in Figure 1).

7.13 REVIEW OF PROBLEMS DETECTABLE BY VIBRATION ANALYSIS

An essential need in specifying effective spectral alarm bands is a firm understanding of what problems are detectable by vibration analysis, how they are detected, and, if detected, how severe they are. Much research has been performed and much continues today on how to evaluate such things as balance condition, alignment, bearing health, gear health, electrical condition, etc. "**Illustrated Vibration Diagnostic Chart**" in Table I represents the best understanding to date of the author on how these problems are best diagnosed, based on much field experience and research of a wide range of articles which have been written on the subject. The list of references researched for this Diagnostic Chart should give an idea of the study which was required to prepare these tables (Reference nos. 2, 3, 4, 5, 6, 9, 11, 12, 15, 16, 17, 20, 21, 22, 23, 25, 26, 27, 29, 30, 31, 34, 35, 36, and 38).

There are several key items included in Table I. First, the plots under "TYPICAL SPECTRUM" column reveal invaluable information about the source of the problem:

1. Which frequencies are present in spectrum and how do they relate to machine operating speed (1X RPM)?
2. What are the amplitudes of each peak?
3. How do the frequency peaks relate to one another? (i.e., "2X RPM is much higher than 1X RPM"; "there is a large peak at 7.43X RPM"; "there are large number of operating speed harmonics present"; "there are high amplitude sidebands around gear mesh frequency"; etc.)

As its column name implies, "TYPICAL SPECTRUM" is meant to be a representative signature for each problem listed in Table I. These spectra are not intended to be all inclusive. For example, referring to "REMARKS" for the Angular Misalignment problem, note that while the typical spectrum shows high amplitude 1X RPM and 2X RPM peaks in the axial direction, the discussion shows that it is not unusual for either 1X, 2X or 3X RPM to dominate the whole spectrum. Similarly, this can occur with either radial misalignment or a cocked bearing. In addition, it is not unusual for a machine to have two or more problems at any one time. For example, if a machine simultaneously had both mechanical looseness and rotor unbalance, they each would be indicated in its spectra which would likely show high 1X RPM, in addition to multiple running speed harmonics.

The next column in Table I is entitled "PHASE RELATIONSHIP". Information on phase is provided for several of the problem sources listed. Amplitude reveals how much something is vibrating. Frequency relates how many cycles occur per unit of time. Phase completes the picture by showing just how the machine is vibrating. Of great importance, phase is a powerful tool in helping differentiate which of several problem sources are dominant. For example, there are a large number of problems that generate vibration at 1X and 2X RPM. Using phase, one learns how the machine is vibrating, and in the process, helps zero in on just which problem is present. For example, Table I shows:

1. Force (or "static") unbalance is evidenced by nearly identical phase in the radial direction on each bearing of a machine (outboard and inboard horizontals are in phase; outboard and inboard verticals are in phase).
2. Couple unbalance shows approximately a 180° out-of-phase relationship when comparing the outboard and inboard horizontal, or outboard and inboard vertical direction phase on the same machine rotor.
3. Angular misalignment is indicated by approximately a 180° phase difference across the coupling, with measurements in the axial direction.

-
4. Parallel misalignment causes radial direction phase across the coupling to be approximately 180° out of phase with respect to one another.
 5. Bent shaft causes axial phase on the same shaft of a machine to approach 180° difference when comparing measurements on the outboard and inboard bearing of the same rotor.
 6. Resonance is shown by exactly a 90° phase change at the point when the forcing frequency coincides with a natural frequency, and approaches a full 180° phase change when the machine passes through the natural frequency (depending on the amount of damping present).
 7. Rotor rub causes significant, instantaneous changes in phase.
 8. Mechanical looseness usually causes phase to be unsteady, with widely differing measurements from one time to the next. The phase measurement may noticeably differ every time you start up the machine, particularly if the rotor itself is loose and rotates on the shaft a few degrees with each startup.

Often, even though phase measurement capability is now offered by most data collectors, users do not use this powerful tool. If not used, this will severely limit the diagnostic capabilities of any program.

Note that "PHASE RELATIONSHIP" is illustrated in each of the first 8 problems of Table I since it is primarily with these problems that phase can be used to differentiate which problem(s) dominate. Phase is then discussed in many of the remarks for the remaining problems in Table I.

Finally, a "REMARKS" column is included in Table I to provide further explanatory information on machine problem symptoms and diagnostics. For example, there is a warning under the remarks column for the "bent shaft" problem source to be sure and account for transducer orientation when taking axial phase measurements.

It is hoped that this diagnostic chart will help users in diagnosing a wide variety of machine problems. Further information is now being researched and field tested which may soon be added to the diagnostic chart.

TABLE I ILLUSTRATED VIBRATION DIAGNOSTIC CHART

PROBLEM SOURCE	TYPICAL SPECTRUM	PHASE RELATIONSHIP	REMARKS
MASS UNBALANCE A. FORCE UNBALANCE			Force Unbalance will be in-phase and steady. Amplitude due to unbalance will increase by the square of speed below first rotor critical (a 3X speed increase = 9X higher vibration). 1X RPM always present and normally dominates spectrum. Can be corrected by placement of only one balance correction weight in one plane at Rotor center of gravity (CG). Approx. 0° phase difference should exist between OB & IB horizontals, as well as between OB & IB verticals. Also, approx. 90° phase difference between horizontal & vertical readings usually occurs on each bearing of unbalanced rotor (±30°).
B. COUPLE UNBALANCE			Couple Unbalance results in 180° out-of-phase motion on same shaft. 1X RPM always present and normally dominates spectrum. Amplitude varies with square of increasing speed below first rotor critical speed. May cause high axial vibration as well as radial. Correction requires placement of balance weights in at least 2 planes. Note that approx. 180° phase difference should exist between OB & IB horizontals, as well as between OB & IB verticals. Also, approx. a 90° difference between the horizontal & vertical phase readings on each bearing usually occurs (±30°).
C. DYNAMIC UNBALANCE			Dynamic Unbalance is the dominant type of unbalance found and is a combination of both force and couple unbalance. 1X RPM dominates the spectrum, and truly requires 2 plane correction. Here, the radial phase difference between outboard and inboard bearings can range anywhere from 0° to 180°. However, the horizontal phase difference should closely match the vertical phase difference, when comparing outboard and inboard bearing measurements (±30°). Secondly, if unbalance predominates, roughly a 90° phase difference usually results between the horizontal and vertical readings on each bearing (±40°).
D. OVERHUNG ROTOR UNBALANCE			Overhung Rotor Unbalance causes high 1X RPM in both Axial and Radial directions. Axial readings tend to be in-phase whereas radial phase readings might be unsteady. However, the horizontal phase differences will usually match the vertical phase differences on the unbalanced rotor (±30°). Overhung rotors have both force and couple unbalance, each of which will likely require correction. Thus, correction weights will most always have to be placed in 2 planes to counteract both force and couple unbalance.
ECCENTRIC ROTOR			Eccentricity occurs when center of rotation is offset from geometric centerline of a pulley, gear, bearing, motor armature, etc. Largest vibration occurs at 1X RPM of eccentric component in a direction thru centerlines of the two rotors. Comparative horizontal and vertical phase readings usually differ either by 0° or by 180° (each of which indicate straight-line motion). Attempts to balance eccentric rotors often result in reducing vibration in one radial direction, but increasing it in the other radial direction (depending on amount of eccentricity).
BENT SHAFT			Bent shaft problems cause high axial vibration with axial phase differences tending towards 180° on the same machine component. Dominant vibration normally occurs at 1X if bent near shaft center, but at 2X if bent near the coupling. (Be careful to account for transducer orientation for each axial measurement if you reverse probe direction.) Use dial indicators to confirm bent shaft.
MISALIGNMENT A. ANGULAR MISALIGNMENT			Angular Misalignment is characterized by high axial vibration, 180° out-of-phase across the coupling. Typically will have high axial vibration with both 1X and 2X RPM. However, not unusual for either 1X, 2X or 3X to dominate. These symptoms may also indicate coupling problems as well. Severe angular misalignment may excite many 1X RPM harmonics. Unlike Mechanical Looseness Type 3, these multiple harmonics do not typically have a raised noise floor on the spectra.
B. PARALLEL MISALIGNMENT			Offset Misalignment has similar vibration symptoms to Angular, but shows high radial vibration which approaches 180° out-of-phase across coupling. 2X often larger than 1X, but its height relative to 1X is often dictated by coupling type and construction. When either Angular or Radial Misalignment becomes severe, they can generate either high amplitude peaks at much higher harmonics (4X-8X), or even a whole series of high frequency harmonics similar in appearance to mechanical looseness. Coupling type and material will often greatly influence the entire spectrum when misalignment is severe. Does not typically have raised noise floor.
C. MISALIGNED BEARING COCKED ON SHAFT			Cocked Bearing will generate considerable axial vibration. Will cause Twisting Motion with approximately 180° phase shift top to bottom and/or side to side as measured in axial direction on same bearing housing. Attempts to align coupling or balance the rotor will not alleviate problem. Bearing usually must be removed and correctly installed.
RESONANCE			Resonance occurs when a Forcing Frequency coincides with a System Natural Frequency, and can cause dramatic amplitude amplification, which might result in premature, or even catastrophic failure. This may be a natural frequency of the rotor, but can often originate from support frame, foundation, gearbox or even drive belts. If a rotor is at or near resonance, it can be almost impossible to balance due to the great phase shift it experiences (90° at resonance; nearly 180° when passes thru). Often requires changing natural frequency to a higher or lower frequency. Natural Frequencies do not generally change with a change in speed which helps facilitate their identification (unless on a large plain bearing machine or on a rotor which has significant overhang).

TABLE I ILLUSTRATED VIBRATION DIAGNOSTIC CHART

PROBLEM SOURCE	TYPICAL SPECTRUM	PHASE RELATIONSHIP	REMARKS
MECHANICAL LOOSENESS	1X RADIAL TYPE A		Mechanical Looseness is indicated by either Type A, B or C vibration spectra. Type A is caused by Structural looseness/weakness of machine feet, baseplate or foundation; also by deteriorated grouting, loose hold-down bolts at the base; and distortion of the frame or base (i.e., soft foot). Phase analysis may reveal approx. 90° to 180° phase difference between vertical measurements on bolt, machine foot, baseplate or base itself.
	5X, 1X, 2X, 3X RADIAL TYPE B		Type B is generally caused by loose pillowblock bolts, cracks in frame structure or in bearing pedestal.
	NOTE RAISED NOISE FLOOR INDICATING LOOSENESS 5X, 1X, 2X, 3X, 4X, 5X, 6X, 7X, 8X, 9X, 10X TYPE C		Type C is normally generated by improper fit between component parts which will cause many harmonics due to nonlinear response of loose parts to dynamic forces from rotor. Causes a truncation of time waveform and a raised noise floor in the spectrum. Type C is often caused by a bearing liner loose in its cap, a bearing loose and turning on its shaft, excessive clearance in either a sleeve or rolling element bearing, a loose impeller on a shaft, etc. Type C Phase is often unstable and may vary widely from one measurement to next, particularly if rotor shifts position on shaft from one startup to next. Mechanical Looseness is often highly directional and may cause very different readings when comparing levels at 30° increments in radial direction all the way around one bearing housing. Also, note that looseness will often cause subharmonic multiples at exactly 1/2 or 1/3X RPM (5X, 1.5X, 2.5X, etc.).
ROTOR RUB	TRUNCATED FLATTENED WAVEFORM 5X, 1X, 2X, 3X, 4X, 5X, 6X, 7X, 8X, 9X, 10X RADIAL		Rotor Rub produces similar spectra to Mechanical Looseness when rotating parts contact stationary components. Rub may be either partial or throughout the entire shaft revolution. Usually generates a series of frequencies, often exciting resonances. Often excites rotor fraction subharmonics of running speed (1/2, 1/3, 1/4, 1/5, ..., 1/n), depending on location of rotor natural frequencies. Rotor rub can excite many high frequencies (similar to wide-band noise when chalk is dragged along a blackboard). It can be very serious and of short duration if caused by shaft contacting bearing babbit. A full annular rub throughout an entire shaft revolution can induce "reverse precession" with the rotor whirling at critical speed in a direction opposite shaft rotation (inherently unstable which can lead to catastrophic failure).
JOURNAL BEARINGS A. WEAR/CLEARANCE PROBLEMS	5X, 1X, 2X, 3X, 4X, 5X, 6X, 7X RADIAL NOTE RAISED NOISE FLOOR INDICATING CLEARANCE/LOOSENESS.		Later stages of journal bearing wear are normally evidenced by presence of whole series of running speed harmonics (up to 10 or 20). Wiped journal bearings often will allow high vertical amplitudes compared to horizontal, but, may show only one pronounced peak at 1X RPM. Journal bearings with excessive clearance may allow a minor unbalance and/or misalignment to cause high vibration which would be much lower if bearing clearances were set to spec.
B. OIL WHIRL INSTABILITY	(.40 - .48 X RPM) 1X RADIAL		Oil Whirl instability occurs at .40 - .48X RPM and is often quite severe. Considered excessive when amplitude exceeds 40% of bearing clearances. Oil Whirl is an oil film excited vibration where deviations in normal operating conditions (attitude angle and eccentricity ratio) cause oil wedge to "push" shaft around within bearing. Destabilizing force in direction of rotation results in a whirl (or forward precession). Oil Whirl is unstable since it increases centrifugal forces which increase whirl forces. Can cause oil to no longer support shaft and can become unstable when whirl frequency coincides with a rotor natural frequency. Changes in oil viscosity, lube pressure and external preloads can affect oil whirl.
C. OIL WHIP INSTABILITY			Oil Whip may occur if machine operated at or above 2X rotor critical frequency. When rotor brought up to twice critical speed, whirl will be very close to rotor critical and may cause excessive vibration that oil film may no longer be capable of supporting. Whirl speed will actually "lock onto" rotor critical and this peak will not pass through it even if machine is brought to higher and higher speeds. Produces a lateral forward precessional subharmonic vibration at rotor critical frequency. Inherently unstable which can lead to catastrophic failure.
ROLLING ELEMENT BEARINGS (4 Failure Stages)	DOMINANT FAILURE SCENARIO ZONE A: BEARING DEFECT BEARING COMPON. FREQ. REGION ZONE B: NATURAL FREQ. REGION ZONE C: SPIKE ENERGY (HFD) ZONE D: SPIKE ENERGY (HFD) STAGE 1: 1X, 2X, 3X STAGE 2: 30K, 120K STAGE 3: BPFI, BPFO, BSF, RTF, 2BPFI, 2BPFO STAGE 4: RANDOM HIGH FREQ. VIBRATION SIDE BAND FREQ. pSE / HFD DECREASES AT FIRST; THEN GROWS SIGNIFICANTLY AT END		4 ROLLING ELEMENT BEARING FAILURE STAGES STAGE 1: Earliest indications of bearing problems appear in ultrasonic frequencies ranging from about 250,000 - 350,000 Hz; later, as wear increases, usually drops to approximately 20,000 - 60,000 Hz (1,200,000 - 3,600,000 CPM). These are frequencies evaluated by Spike Energy (gSE), HFD(g) and Shock Pulse (dB). For example, spike energy may first appear at about .25 gSE in Stage 1 (actual value depending on measurement location and machine speed). Acquiring high frequency enveloped spectra confirms whether or not bearing is in Failure Stage 1. STAGE 2: Slight bearing defects begin to "ring" bearing component natural frequencies (f _n) which predominantly occur in 30K - 120K CPM range. Such natural frequencies may also be resonances of bearing support structures. Sideband frequencies appear above and below natural frequency peak at end of Stage 2. Overall spike energy grows (for example, from .25 to .50 gSE). STAGE 3: Bearing defect frequencies and harmonics appear. When wear progresses, more defect frequency harmonics appear and number of sidebands grow, both around these and bearing component natural frequencies. Overall spike energy continues to increase (for example, from .5 to over 1 gSE). Wear is now usually visible and may extend throughout periphery of bearing, particularly when many well formed sidebands accompany bearing defect frequency harmonics. High frequency demodulated and enveloped spectra help confirm Stage III. Replace bearings now! (Independent of bearing defect frequency amplitudes in vibration spectra). STAGE 4: Towards the end, amplitude of 1X RPM is even affected. It grows, and normally causes growth of many running speed harmonics. Discrete bearing defect and component natural frequencies actually begin to "disappear" and are replaced by random, broadband high frequency "noise floor". In addition, amplitudes of both high frequency noise floor and spike energy may in fact decrease; but just prior to failure, spike energy and HFD will usually grow to excessive amplitudes.

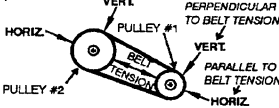
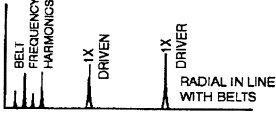
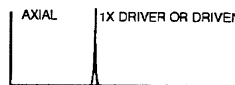
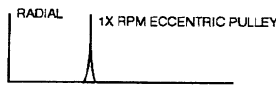
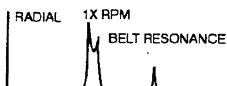
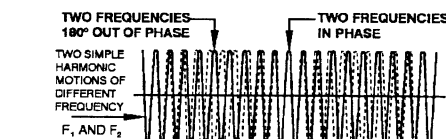
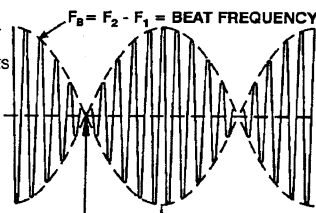
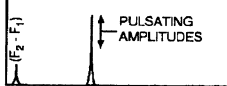
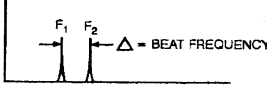
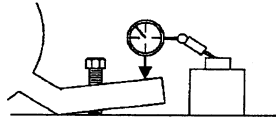
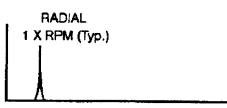
TABLE I ILLUSTRATED VIBRATION DIAGNOSTIC CHART

PROBLEM SOURCE	TYPICAL SPECTRUM	REMARKS
HYDRAULIC AND AERODYNAMIC FORCES		
A. BLADE PASS & VANE PASS	BPF = # BLADES X RPM	Blade Pass Frequency (BPF) = No. of Blades (or Vanes) X RPM. This frequency is inherent in pumps, fans and compressors, and normally does not present a problem. However, large amplitude BPF (and harmonics) can be generated in pump if gap between rotating vanes and stationary diffusers is not equal all the way around. Also, BPF (or harmonic) sometimes can coincide with a system natural frequency causing high vibration. High BPF can be generated if impeller wear ring seizes on shaft, or if welds fastening diffuser vanes fail. Also, high BPF can be caused by abrupt bends in pipe (or duct), obstructions which disturb flow, damper settings or if pump or fan rotor is positioned eccentrically within housing.
B. FLOW TURBULENCE	BPF = BLADE OR VANE PASS FREQUENCY	Flow Turbulence often occurs in blowers due to variations in pressure or velocity of the air passing thru the fan or connected ductwork. This flow disruption causes turbulence which will generate random, low frequency vibration, typically in the range of 50 to 2000 CPM. If surging occurs within a compressor, random broadband high frequency vibration can occur. Excessive turbulence can also excite broadband high frequency.
C. CAVITATION	120K CPM	Cavitation normally generates random, higher frequency broadband energy which is sometimes superimposed with blade pass frequency harmonics. Normally indicates insufficient suction pressure (starvation). Cavitation can be quite destructive to pump internals if left uncorrected. It can particularly erode impeller vanes. When present, it often sounds as if "gravel" is passing thru pump. Cavitation is usually caused by insufficient inlet flow. Can occur during one survey, and be absent the next survey (if changes in suction valve settings are made).
GEARS		
A. NORMAL SPECTRUM	$GMF = \#T_g \times RPM_g = \#T_p \times RPM_p$	Normal Spectrum shows Gear & Pinion Speeds, along with Gear Mesh Frequency (GMF) and very small GMF harmonics. GMF harmonics commonly will have running speed sidebands around them. All peaks are of low amplitude, and no natural frequencies of gears are excited. F_{max} recommended at 3.25X GMF (minimum) when # teeth are known. If tooth count is not known, set F_{max} at 200X RPM on each shaft.
B. TOOTH WEAR	51T, 26T	Key indicator of Tooth Wear is excitation of Gear Natural Frequency (f_n), along with sidebands around it spaced at the running speed of the bad gear. Gear Mesh Frequency (GMF) may or may not change in amplitude, although high amplitude sidebands and number of sidebands surrounding GMF usually occur when wear is noticeable. Sidebands may be better wear indicator than GMF frequencies themselves. Also, high amplitudes commonly occur at either 2XGMF or at 3XGMF (esp. 3XGMF), even when GMF amplitude is acceptable.
C. TOOTH LOAD		Gear Mesh Frequencies are often very sensitive to load. High GMF amplitudes do not necessarily indicate a problem, particularly if sideband frequencies remain low level, and no gear natural frequencies are excited. Each Analysis should be performed with system at maximum operating load for meaningful spectral comparisons.
D. GEAR ECCENTRICITY AND BACKLASH		Fairly high amplitude sidebands around GMF harmonics often suggest gear eccentricity, backlash, or non-parallel shafts which allow the rotation of one gear to "modulate" either the GMF amplitude or the running speed of the other gear. The gear with the problem is indicated by the spacing of the sideband frequencies. Also, 1X RPM level of eccentric gear will normally be high if eccentricity is the dominant problem. Improper backlash normally excites GMF harmonics and Gear Natural Frequency, both of which will be sidebanded at 1X RPM. GMF amplitudes will often decrease with increasing load if backlash is the problem.
E. GEAR MISALIGNMENT		Gear Misalignment almost always excites second order or higher GMF harmonics which are sidebanded at running speed. Often will show only small amplitude 1X GMF, but much higher levels at 2X or 3X GMF. Important to set F_{max} high enough to capture at least 3 GMF harmonics. Also, sidebands around 2XGMF will often be spaced at 2X RPM. Note that sideband amplitudes often are not equal on left and right side of GMF and GMF harmonics due to the tooth misalignment. Causes uneven wear pattern.
F. CRACKED/BROKEN TOOTH	TIME WAVEFORM $\Delta = \frac{1}{RPM}$ OF GEAR WITH BROKEN OR CRACKED TOOTH	A Cracked or Broken Tooth will generate a high amplitude at 1X RPM of this gear only in the time waveform, plus it will excite gear natural frequency (f_n) sidebanded at its running speed. It is best detected in Time Waveform which will show a pronounced spike every time the problem tooth tries to mesh with teeth on the mating gear. Time between impacts (Δ) will correspond to 1/RPM of gear with the problem. Amplitudes of Impact Spikes in Time Waveform often will be 10X to 20X higher than that at 1X RPM in the FFT!
G. GEAR ASSEMBLY PHASE PROBLEMS	$GAPF = \frac{GMF}{N_A}$ $GAPF = 3000 \text{ CPM} = 0.20X \text{ GMF (FRACTIONAL GMF)}$	Gear Assembly Phase Freq. (GAPF) can result in Fractional Gear Mesh Frequencies (if $N_A > 1$). It literally means (T_g/N_A) gear teeth will contact (T_p/N_A) pinion teeth and will generate N_A wear patterns, where N_A in a given tooth combination equals the product of prime factors common to the number of teeth on the gear and pinion (N_A = Assembly Phase Factor). GAPF (or harmonics) can show up right from the beginning if there were manufacturing problems. Also, its sudden appearance in a periodic survey spectrum can indicate damage if contaminant particles pass through the mesh, resulting in damage to the teeth in mesh at the time of ingestion just as they enter and leave meshing or that gears have been reoriented.
H. HUNTING TOOTH PROBLEMS	$f_{HT} = \frac{(GMF)(N_A)}{(T_{GEAR})(T_{PINION})}$ $N_A = 1$ is the ideal assembly phase factor in gear design $f_{HT} = \frac{(6 \times 1000)(1)}{(6)(7)} = 143 \text{ CPM (One Pulse Per 7 Pinion Revolutions)}$	Hunting Tooth Frequency (f_{HT}) occurs when faults are present on both the gear and pinion which might have occurred during the manufacturing process, due to mishandling, or in the field. It can cause quite high vibration, but since it occurs at low frequencies predominately less than 600 CPM, it is often missed. A gear set with this tooth repeat problem normally emits a "growing" sound from the drive. The maximum effect occurs when the faulty pinion and gear teeth both enter mesh at the same time (on some drives, this may occur only 1 or every 10 to 20 revolutions, depending on the f_{HT} formula). Note that T_{GEAR} and T_{PINION} refer to number of teeth on the gear and pinion, respectively. N_A is the Assembly Phase Factor defined above. Will often modulate both GMF and Gear RPM peaks.

TABLE I ILLUSTRATED VIBRATION DIAGNOSTIC CHART

PROBLEM SOURCE	TYPICAL SPECTRUM	REMARKS
GEARS (CONTINUED)		
I. LOOSE BEARING FIT		Excessive Clearance of bearings supporting the gears can not only excite many running speed harmonics, but will often cause high amplitude response at GMF, 2GMF and/or 3GMF. These high GMF amplitudes are actually a response to, and not the cause of, looseness within the bearings supporting the gearing. Such excessive clearance can be caused either by extensive bearing wear or by improper bearing fit onto the journal during installation. Left uncorrected, it can cause excessive gear wear and damage to other components.
AC INDUCTION MOTORS		
A. STATOR ECCENTRICITY, SHORTED LAMINATIONS OR LOOSE IRON		Stator problems generate high vibration at 2X line frequency (2FL). Stator eccentricity produces uneven stationary air gap between rotor and stator which produces very directional vibration. Differential Air Gap should not exceed 5% for induction motors and 10% for synchronous motors. Soft foot and warped bases can produce an eccentric stator. Loose iron is due to stator support weakness or looseness. Shorted stator laminations can cause uneven, localized heating which can distort the stator itself. This produces thermally-induced vibration which can significantly grow with operating time causing stator distortion and static air gap problems.
B. ECCENTRIC ROTOR (Variable Air Gap) F_e = Electrical Line Freq. N_s = Synch. Speed = $\frac{120F_e}{P}$ F_s = Slip Freq. = $N_s - \text{RPM}$ F_p = Pole Pass Freq. = $F_e \times P$ P = #Poles		Eccentric Rotors produce a rotating variable air gap between the rotor and stator which induces pulsating vibration (normally between 2FL and closest running speed harmonic). Often requires "zoom" spectrum to separate 2FL and running speed harmonic. Eccentric rotors generate 2FL surrounded by Pole Pass frequency sidebands (F_p), as well as F_s sidebands around running speed. F_p appears itself at low frequency (Pole Pass Frequency = Slip Frequency X #Poles). Common values of F_p range from about 20 to 120 CPM (0.3 - 2.0 Hz). Soft foot or misalignment often induces a variable air gap due to distortion (actually a mechanical problem; not electrical).
C. ROTOR PROBLEMS		Broken or Cracked rotor bars or shorting rings; bad joints between rotor bars and shorting rings; or shorted rotor laminations will produce high 1X running speed vibration with pole pass frequency sidebands (F_p). In addition, these problems will often generate F_s sidebands around the second, third, fourth and fifth running speed harmonics. Loose or open rotor bars are indicated by 2X line freq. ($2FL$) sidebands surrounding Rotor Bar Pass Frequency (RBPF) and/or its harmonics (RBPF = Number of Bars X RPM). Often will cause high levels at 2X RBPF, with only a small amplitude at 1X RBPF. Electrically induced arcing between loose rotor bars and end rings will often show high levels at 2X RBPF (with $2FL$ sidebands); but little or no increase in amplitudes at 1X RBPF.
D. PHASING PROBLEM (Loose Connector)		Phasing problems due to loose or broken connectors can cause excessive vibration at 2X Line Freq. ($2FL$) which will have sidebands around it spaced at 1/3 Line Freq. ($1/3 F_e$). Levels at $2FL$ can exceed 1.0 in/sec if left uncorrected. This is particularly a problem if the defective connector is only sporadically making contact. Loose or broken connectors must be repaired to prevent catastrophic failure.
AC SYNCHRONOUS MOTORS		
(Loose Stator Coils)		Loose stator coils in synchronous motors will generate fairly high vibration at Coil Pass Freq. (CPF) which equals the number of stator coils X RPM (#Stator Coils = #Poles X #Coils/Pole). The Coil Pass Frequency will be surrounded by 1X RPM sidebands. Synchronous motor problems may also be indicated by high amplitude peaks at approx. 60,000 to 90,000 CPM, accompanied by $2FL$ sidebands. Take at least one spectrum up to 90,000 CPM on each motor bearing housing.
DC MOTORS AND CONTROLS		
A. NORMAL SPECTRUM		Many DC Motor and Control Problems can be detected by vibration analysis. Full-wave rectified motors (6 SCR's) generate a signal at 6X Line Frequency ($6FL = 360 \text{ Hz} = 21,600 \text{ CPM}$); while half-wave rectified DC motors (3 SCR's) generate 3X Line Freq. ($3FL = 180 \text{ Hz} = 10,800 \text{ CPM}$). The SCR firing Frequency is normally present in a DC Motor Spectrum, but at low amplitude. Note the absence of other peaks at multiples of F_e .
B. BROKEN ARMATURE WINDINGS, GROUNDING PROBLEMS OR FAULTY SYSTEM TUNING		When DC Motor spectra are dominated by high levels at SCR or 2X SCR, this normally indicates either Broken Motor Windings or Faulty Tuning of the Electrical Control System. Proper tuning alone can lower vibration at SCR and 2X SCR significantly if control problems predominate. High amplitudes at these frequencies would normally be above approximately .10 in/sec, peak at 1X SCR and about .04 in/sec at 2X SCR Firing Freq.
C. FAULTY FIRING CARD OR BLOWN FUSE		When one firing card fails to fire, then 1/3 of power is lost, and can cause repeated momentary speed changes in the motor. This can lead to high amplitudes at 1/3X and 2/3X SCR Frequency ($1/3X \text{ SCR Freq.} = 1X F_e$ for half-wave rectified, but $2X F_e$ for a full-wave rectified SCR). Caution: Card/SCR configuration should be known before troubleshooting motor (#SCR's, #Firing Cards, etc.).
D. FAULTY SCR, SHORTED CONTROL CARD, LOOSE CONNECTIONS AND/OR BLOWN FUSE		Faulty SCR's, Shorted Control Cards and/or Loose Connections can generate noticeable amplitude peaks at many combinations of line frequency (F_e) and SCR firing frequency. Normally, 1 bad SCR can cause high levels at F_e and/or $5FL$ in 6 SCR motors. The point to be made is that neither F_e , $2FL$, $4FL$, nor $5FL$ should be present in DC Motor spectra.
E. FAULTY COMPARATOR CARD		Faulty Comparator Cards cause problems with RPM fluctuation or "hunting". This causes a constant collapsing and regenerating of the magnetic field. These sidebands often approximate the RPM fluctuation and require a high resolution FFT to even detect them. Such sidebands could also be due to generation and regeneration of the magnetic field.
F. ELECTRICAL CURRENT PASSAGE THRU DC MOTOR BEARINGS		Electrically-induced Fluting is normally detected by a series of difference frequencies with the spacing most often at the outer race defect frequency (BPFO), even if such fluting is present on both the outer and inner races. They most often show up in a range centered at about 100,000 to 150,000 CPM. A 180K CPM spectrum with 1600 lines is recommended for detection with measurements on both the OB and IB DC motor bearings.

TABLE I ILLUSTRATED VIBRATION DIAGNOSTIC CHART

PROBLEM SOURCE	TYPICAL SPECTRUM	REMARKS
BELT DRIVE PROBLEMS A. WORN, LOOSE OR MISMATCHED BELTS   PITCH DIAM. X RPM₁ = PITCH DIAM. X RPM₂ BELT FREQ. = $3.142 \times \text{PULLEY RPM} \times \text{PITCH DIAM.}$ BELT LENGTH TIMING BELT FREQ. = BELT FREQ. X #BELT TEETH = PULLEY RPM X #PULLEY TEETH Belt frequencies are below the RPM of either the motor or the driven machine. When they are worn, loose or mismatched, they normally cause 3 to 4 multiples of belt frequency. Often 2X belt freq. is the dominant peak. Amplitudes are normally unsteady, sometimes pulsing with either driver or driven RPM. On timing belt drives, wear or pulley misalignment is indicated by high amplitudes at the Timing Belt Frequency. Chain drives will indicate problems at Chain Pass Frequency which equals #Sprocket Teeth X RPM.		
B. BELT/PULLEY MISALIGNMENT   Misalignment of pulley produces high vibration at 1X RPM predominantly in the axial direction. The ratio of amplitudes of driver to driven RPM depends on where the data is taken, as well as on relative mass and frame stiffness. Often with pulley misalignment, the highest axial vibration on the motor will be at fan RPM, or vice versa. Can be confirmed by phase measurements by setting Phase Filter at RPM of pulley with highest axial amplitude; then compare phase at this particular frequency on each rotor in the axial direction.		
C. ECCENTRIC PULLEYS   Eccentric pulleys cause high vibration at 1X RPM of the eccentric pulley. The amplitude is normally highest in line with the belts, and should show up on both driver and driven bearings. It is sometimes possible to balance eccentric pulleys by attaching washers to taper-lock bolts. However, even if balanced, the eccentricity will still induce vibration and reversible fatigue stresses in the belt. Pulley eccentricity can be confirmed by phase analysis showing horizontal & vertical phase differences of nearly 0° or 180°.		
D. BELT RESONANCE   Belt Resonance can cause high amplitudes if the belt natural frequency should happen to approach, or coincide with, either the motor or driven RPM. Belt natural frequency can be altered by changing either the belt tension, belt length or cross section. Can be detected by tensioning and then releasing belt while measuring the response on pulleys or bearings. However, when operating, belt natural frequencies will tend to be slightly higher on the tight side and lower on the slack side.		
BEAT VIBRATION   Two simple harmonic motions of different frequency, F_1 and F_2 $F_B = F_2 - F_1 = \text{BEAT FREQUENCY}$ BEAT FREQUENCY GENERATED BY TWO FREQUENCIES ABOVE MINIMUM VIBRATION OCCURS WHEN 2 FREQUENCIES ARE 180° OUT OF PHASE MAXIMUM VIBRATION OCCURS WHEN 2 FREQUENCIES ARE IN PHASE A Beat Frequency is the result of two closely spaced frequencies going into and out of synchronization with one another. The wideband spectrum normally will show one peak pulsating up and down. When you zoom into this peak (lower spectrum below), it actually shows two closely spaced peaks. The difference in these two peaks ($F_2 - F_1$) is the beat frequency which appears itself in the wideband spectrum. The beat frequency is not commonly seen in normal frequency range measurements since it is inherently low frequency, usually ranging from only approximately 5 to 100 CPM. Maximum vibration will result when the time waveform of one frequency (F_1) comes into phase with the waveform of the other frequency (F_2). Minimum vibration occurs when waveforms of these two frequencies line up 180° out of phase.   WIDEBAND SPECTRUM ZOOM SPECTRUM		
SOFT FOOT, SPRUNG FOOT AND FOOT-RELATED RESONANCE   "Soft Foot" occurs when a machine's foot or frame deflects greatly when a hold-down bolt is loosened to hand tightness, causing the foot to rise more than approximately .002 - .003 inch. This does not always cause a great vibration increase. However, it can do so if the soft foot affects alignment or motor air gap concentricity. "Sprung Foot" can cause great frame distortion, resulting in increased vibration, force and stress in the frame, bearing housing, etc. This can occur when a hold-down bolt is forcefully torqued down on the sprung foot in an attempt to level the foot. "Foot-Related Resonance" can cause dramatic amplitude increases from 5X to 15X or more, as compared with that when the bolt (or combination of bolts) is loosened to hand tightness. When tight, this bolt can notably change the natural frequency of the foot or machine frame itself. Soft Foot, Sprung Foot or Foot-Related Resonance most often affects vibration at 1X RPM, but can also do so at 2X RPM, 3X RPM, 2X line frequency, blade pass frequency, etc. (particularly Foot-Related Resonance).		

7.14 SPECIFICATION OF OVERALL VIBRATION ALARM LEVELS AND EXPLANATION OF THE ORIGIN OF TABLE II "OVERALL CONDITION RATING" CHART

Much work continues today on establishing standards for allowable overall vibration. Various national and international committees made up of experienced professionals have been established and are given the charge of formulating these vibration criteria. This includes the international working group on machinery vibration standards which is now working to update several criteria (19):

- ISO 2372 - "Mechanical Vibration of Machines with Operating Speeds from 10 to 200 Revolutions per Second" - Basis for specifying evaluation standards (measurements made on structure).
- ISO 3945 - "Mechanical Vibration of Large Rotating Machines with Speeds Ranging from 10 to 200 Revolutions per Second" - Measurement and evaluation of vibration severity in situ (measurements made on structure at various elevations).
- ISO 7919 - "Mechanical Vibration of Non-Reciprocating Machines" - Measurement on rotating components and evaluation (measurements made on shafting).

Some attempts have been made, and some are now being offered, to provide vibration criteria based on the type of machine and its drive configuration (centrifugal pump, direct coupled fan, belt driven fan, turbine/generator, etc.), and on its mounting (isolated versus non-isolated). It is recognized that there is often dramatic difference in the amount of vibration from one machine type versus another. For example, a reciprocating air compressor obviously has significantly more inherent vibration than does a hermetic chiller or, for that matter, a precision machine tool spindle. Also, a machine will generally experience higher vibration than before when placed on isolators, depending on the isolator type, isolator connection, isolator natural frequency, forcing frequencies of the machine itself, machine center of gravity relative to the placement of isolators, etc. Thus, it is important that the user of today's condition monitoring hardware and software take into account both the type of machine and its mounting when he begins to specify alarm levels of overall vibration for each machine that he will input into his computer database.

In addition, it is important for the user to know how his particular predictive maintenance data collector and software system measures overall vibration. Some systems have a fixed frequency range, completely independent of any frequency range chosen on any particular spectrum. In fact, this overall measurement is completely independent from spectral measurements in some systems. In others, the overall is determined by first taking a spectrum, and then by using Equation (1) to calculate the spectral overall level [Equation (1) which is repeated here for the reader]:

$$OA = \frac{\sqrt{\sum_{i=1}^n A_i^2}}{\sqrt{N_{BF}}} = \frac{\sqrt{A_1^2 + A_2^2 + A_3^2 + \dots + A_N^2}}{\sqrt{1.5}}$$

Where:

OA = Overall Level of Spectrum or Band
n = Number of FFT Lines of Resolution
A_i = Amplitude of Each of the FFT Lines
N_{BF} = Noise Bandwidth for Window Chosen
1.5 for Hanning Window

$$\text{Which means } OA = .8165 \sqrt{\sum_{i=1}^n A_i^2}$$

During the years, our company has had the opportunity of analyzing a diverse array of both process and utility machinery ranging from very small, precision, high-speed spindles to large, slow moving machines. In addition, this work has been performed for a wide array of industry types. This has given us invaluable exposure which has been greatly beneficial when opportunities have arisen for setting up condition monitoring programs for these same clients.

Through the years, we have developed in-house vibration criteria specifically for the purpose of setting up these condition monitoring databases. Some of the criteria we have developed is included in Table II. Note that Table II includes overall peak velocity criteria for measurements taken on the machine structure. Importantly, levels specified in Table II are not meant to be final, concrete numbers, but are intended to be a starting point when nothing is known about a machine other than its nameplate data, machine type and mounting. Later, after taking actual measurements on each point of each machine, these levels are individually reviewed and adjusted as needed. This refinement procedure is discussed later in the paper and examples are given illustrating the procedure.

Note that each of four "ratings" are provided in Table II including "GOOD", "FAIR", "ALARM 1" and "ALARM 2". After review of all spectra captured on a machine, if no problems are found, the first two columns ("GOOD" and "FAIR") are offered to give the client a general feel for the overall condition of each machine based on the highest overall level measured on his machine. However, even if the highest overall on a machine might remain within the "GOOD" range, it is still possible for the machine to be in alarm, depending on what frequencies were generated, the amplitudes of those frequencies, and the problem source(s) generating these frequencies. That is where the spectral alarm bands come into play to ferret out the "apparently good condition" machines from those that truly have a problem. Corrective actions should be taken on those machines having vibration exceeding "ALARM 1"; while those exceeding "ALARM 2" are felt to be exposed to such high levels as to render potentially catastrophic failure (therefore, demanding immediate attention). Amplitudes listed in Table II were developed by calculating both the average level and standard deviation of large quantities of diverse types of machinery over a period of approximately 20 years in carrying out condition monitoring programs. Then, "ALARM 1" overall levels were calculated by summing the average level plus 3 times the standard deviation [see Equations (2) through (4B)]. Final statistical overall levels were then rounded to the nearest ".025" level (that is, a level of .318 would be rounded off to .325 in/sec). Finally "ALARM 2" levels were determined by increasing "ALARM 1" levels by 50%.

Importantly, not only do the overall levels specified in Table II serve as an overall alarm given in the PMP database, but also they are used as direct input for specifying alarm levels for each specific spectral band in the section which will follow. Of course, if this overall is later refined after making several measurements on a machine or on a group of machines, the spectral band alarm levels themselves will also have to be adjusted as well.

7.15 SPECIFICATION OF SPECTRAL ALARM LEVELS AND FREQUENCY BANDS USING TABLE III

Table III provides the tabulated procedure on how to originally specify spectral alarm bands for various machine types and configurations using those types of condition monitoring software systems which allow the spectrum to be broken up into 6 individual bands. Each of these bands can be set at any span of frequencies, and at any alarm level for each individual band as chosen by the user. Therein lies both the strength of spectral alarm bands, and paradoxically, their major weakness if the user himself does not know where each frequency span should be specified, nor how high to set each one of the band alarm levels. Therefore, the express purpose of this section is to provide the condition monitoring software user with the capability, not only of originally setting up a PMP database using proper spectral alarm bands, but also to help him refine his database on which he might have been taking measurements for several years. Several years ago, our company made a detailed in-house study on how to specify these bands. At the conclusion

of this study, we elected to develop a written, tabular procedure on how to properly specify them. Since that time, we have helped a number of clients set up bands on their specific machinery in their particular industry. In so doing, we have continued to learn more and more about how to best use them, and have "polished" and refined our techniques several times. In addition, much study went into preparation of Table III as can be seen by the list of references (see Reference nos. 3, 6, 8, 10, 11, 12, 13, 14, 18, 23, 26, 28, 29, 32, and 37).

Importantly, please note that the procedures specified in Table III assume casing measurements of peak velocity (in/sec) using instruments which measure RMS and "convert" them to peak levels by electronic multiplication of amplitudes by 1.414. This now includes most of the data collectors in use in the United States. Also, Table III specifies spectral bands whose alarm levels are compared to the total power within the band (so-called "Power Bands"). Please refer to the section entitled "Two Types of Spectral Alarm Bands". Although Table III applies to peak velocity amplitudes, the reader can modify it for RMS simply by multiplying amplitudes by .707. Then, if he wishes to have them expressed in metric units (mm/sec), he can multiply these RMS in/sec amplitudes by a factor of 25.4 and rounding them to the nearest appropriate metric level.

Table III shows how spectral alarm bands are set up for a number of machine types and configurations. Cases A and B are for both the driver and driven components of general rotating machines which are outfitted with rolling element and sleeve bearings, respectively. Cases C and D specify high frequency measurement points which are to be taken on gearbox housings in close proximity to each gear mesh, and which are essential to evaluate the health and alignment of gearing. Case C assumes one knows the number of gear teeth, while Case D shows how to specify alarm bands for gearboxes where the number of teeth is unknown. Cases E and F are special points with the purpose of detecting potential motor electrical problems. The point specified by Case E is intended to detect the first and second harmonic rotor bar pass frequencies (number of bars X RPM), whereas the point specified by Case F attempts to separate mechanical from electrical vibration sources, particularly in the vicinity of machine operating speed, electrical synchronous frequency (60 Hz), and twice synchronous frequency (120 Hz). Case G covers how to specify alarm bands for centrifugal compressors, blowers and pumps. Cases H and I have been added to this paper in its fourth edition. Case H covers DC motors and controls while Case I encompasses machine tool spindles.

Importantly, the specification procedure outlined in Table III applies to general process and utility machinery such as centrifugal pumps, blowers, motors, forced-draft fans, induction-draft fans, motor/generator sets, centrifugal air compressors, refrigeration chillers, vacuum pumps, boiler feed pumps, gearboxes, etc. These specs do not apply to more specialized machine types such as reciprocating or rotary screw compressors; diesel engines; gas turbines; large turbine/generators; exciters; lobe-type rotary blowers; pulverizers; etc. Normally, spectral bands for these machine types have to be "custom-designed" for each set or grouping of them, and even then, often require the capture of several sets of data before one can begin to establish meaningful alarm bands.

For example, lobe-type rotary blowers (i.e., "Roots Blowers") present a real problem to the user who attempts to specify one all-encompassing set of alarm bands. They are offered in a wide range of sizes and configurations. Often, even after several surveys are conducted on these machines, the user may have difficulty in adequately specifying alarm bands since even identically sized and driven rotary blowers still can exhibit unique sets of vibration spectra (23). In reality, only 6 spectral alarm bands cannot adequately address these machines. They need approximately 10 to 12 bands (or more) to adequately cover them.

However, if the user is given the assignment of specifying spectral alarm bands for his plant, either when originally setting up its database or after several years of data have been captured (without adequate alarm bands), the procedure given in Table III should cover a large percentage of his machines. Before entering Table III, the user should identify his particular machine type and refer

to Table II to find the alarm level of overall vibration for this machine. This will be used as direct input into the spectral alarm band specs of Table III. If his particular machine type is not included in Table II, the user should either refer to the manufacturer of his machine, other similar vibration severity charts, or use alarm levels for another machine type listed in Table II which most closely resembles his machine.

Please refer to the entries under the first column of Table III. The "BAND LOWER FREQUENCY" specifies at what frequency each band should begin, whereas "BAND HIGHER FREQUENCY" shows where each band should end (for example, "from 60 to 1000 CPM"). In general, no gaps should be left between bands, nor should bands overlap one another (although some analysts using power bands sometimes extend one band from the beginning to the end of a complete spectrum in order to have the system calculate the "Spectral Overall Level", and then compare this to the overall level provided separately by their instrument). Next, the column entry entitled "BAND ALARM" specifies how high to set the alarm level of each band.

Notice that many of the cases described in Table III have the "BAND LOWER FREQUENCY" set at 1% F_{MAX} rather than at 0 CPM. The reason for this is that data collectors and spectrum analyzers most always have built-in "noise" within the first 1 to 3 FFT lines, particularly when data from an accelerometer are electronically integrated to velocity. In fact, some instruments have been known to display "peaks" with so-called "amplitudes" over 2.0 in/sec within these first 3 FFT lines. If F_{MAX} is properly specified, the first 2 to 3 FFT lines will almost always be contaminated with such electronic and/or integration "noise". Therefore, Band 1 will never begin within these first 3 lines in Table III.

Each of the cases specify the maximum frequency (F_{MAX}) which is always given along with the case title. Therefore, each case will tell where to set both the frequency range and alarm level of each band, and will describe what each band covers (i.e., bearing defect frequencies, gear mesh frequencies, etc.). Case A will be discussed in detail to illustrate the alarm band specification technique, whereas only highlights of each remaining case will be given. Then, several examples will follow the discussions to further illustrate how these techniques should be applied.

**Case A - General Rolling Element Bearing Machine Without Rotating Vanes:
(Motors, Gearbox Lower Frequency Measurements, etc.)**

Case A applies to a wide range of general rotating process and utility machines which are outfitted with rolling element bearings (ball, roller or needle bearings). Before entering Table III, refer to Table II to obtain the alarm level of overall peak velocity for your machine type. Then, determine the type of rolling element bearing. For common rolling element bearings, Case A specifies a spectrum with a maximum frequency (F_{MAX}) of approximately 50X RPM (for example, for a nominal speed of 1800 RPM, set F_{MAX} at approximately 90,000 CPM). However, for tapered roller bearings (Timken cup and cone arrangement, or equivalent) or for spherical roller bearings, Case A specifies a maximum frequency of approximately 60X RPM. The reason for the higher F_{MAX} for these bearing types is the fact that, with their particular geometries, they inherently have higher calculated rolling element bearing defect frequencies. Also note that if the speed is below 1700 RPM, F_{MAX} must be set higher than 50X RPM (as seen in notes of Case A). The reason for this is to ensure that the spectra designed for this machine will detect a rolling element bearing in only the second of four failure stages through which it will normally pass rather than waiting late in the life of the bearing before problems are detected. Referring to Table I for "Rolling Element Bearings", note that the natural frequencies of bearing components will be excited during this second stage. Since these natural frequencies normally range from 30,000 to 90,000 CPM for most bearings, it is important to keep F_{MAX} sufficiently high to detect these when excited (bearing natural frequencies may range as high as 120,000 to 150,000 for specialty rolling element bearings such as aircraft bearings, small bore bearings, etc.).

Please note that it is not necessary to specify F_{MAX} at exactly 50X or 60X RPM, but it should be somewhere in this vicinity (certainly not less than 45X RPM). If one sets F_{MAX} too low, it can cause a spectrum to completely miss potentially serious developing bearing wear, particularly during earlier stages. On the other hand, if F_{MAX} is set too high, this can result in poor frequency resolution which can cause the user to misdiagnose problems, since he does not have sufficiently precise frequency resolution to properly identify such frequency components as true running speed harmonics versus bearing defect frequencies, or vibration transmitting from adjacent machines. Also, if one sets F_{MAX} too high, this can cause potentially valuable information on subsynchronous vibration to be "buried" on the left-hand side of the spectrum. In general, the rule of thumb is to keep F_{MAX} as low as you can "without missing anything important".

Referring to Case A in Table III, note that each one of the bands has a specific purpose and zone of coverage. For example, Band 1 ranges from subsynchronous vibration (below 1X RPM) up through operating speed. Bands 2 and 3 cover 2X and 3X RPM, respectively. Band 4 will include fundamental bearing defect frequencies for most rolling element bearings. Similarly, Bands 5 and 6 will include bearing defect frequency harmonics, as well as natural frequencies of bearing components for most common rolling element bearings.

Now, referring back to Table III, note that Band 1 extends from 1% of F_{MAX} to a frequency at 1.2X RPM. In the case of the example 1800 RPM machine shown in Figure 10 having F_{MAX} at 90,000 CPM, Band 1 would extend from 900 to 2160 CPM. The Band 1 alarm spec calls for 90% of the overall level. Thus, if the overall alarm were .300 in/sec (from Table II), then the Band 1 alarm would be set at .270 in/sec for this machine. Similarly, Table III specifies the frequency range of Band 2 to extend from 1.2 to 2.2X RPM (in the 1800 RPM case, this would extend from approximately 2160 to 3960 CPM). The Band 2 alarm spec calls for 30% of the overall alarm (thus for the example .300 in/sec overall, Band 2 would be set at .090 in/sec). Finally, Bands 3 through 6 are specified similarly.

Case B - General Sleeve Bearing Machine Without Rotating Vanes: (Sleeve Bearing Motors, Gearbox Lower Frequency Measurements, etc.)

Case B is similar to Case A, but is for general machines outfitted with sleeve bearings. Incidentally, if a sleeve bearing motor is driving a rolling element machine, Case A (rolling element) would be used for the driven machine points, whereas Case B (sleeve bearing) would be applied to the points on the motor. However, refer to Cases G thru I if the driven component is a centrifugal machine, DC motor or machine tool spindle.

Notice that F_{MAX} for these sleeve bearing machines is set only at 20X RPM as compared to 50X up to 120X RPM on rolling element bearing machines which inherently have much higher frequency spectra. In addition, some potentially serious problems can occur at subsynchronous frequencies on sleeve bearings including such things as oil whirl and oil whip. Therefore, this subsynchronous band needs to have good frequency resolution and must be closely watched.

Band 1 covers only the subsynchronous vibration in Case B while Bands 2, 3 and 4 include 1X, 2X and 3X RPM peaks, respectively (see Table III). Band 5 covers the range from 4X through 10X RPM while Band 6 extends from 10.5X RPM to F_{MAX} . Here again, the highest alarm level specified for any of the bands in Case B will be that at 1X RPM (Band 2). On the other hand, little amplitude is allowed in Band 6 even though it covers about 50% of the entire spectrum since only insignificant vibration should occur in this region if problems are not present, particularly if this machine is not a gearbox or connected to a gearbox.

Case C - Gearbox High Frequency Points with Known Number of Teeth:

Gearboxes require two sets of measurements on the same points due to the fact that many gear problems are detected at very high frequencies as compared to vibration due to such problems as

unbalance, misalignment, etc. Therefore, one set of measurements should be taken on the gearbox using either Case A or B, depending on whether the gearbox is outfitted with rolling element (Case A) or sleeve (Case B) bearings. Then, a second set of measurements should be taken at various gearbox points close to each mesh, with F_{MAX} on this second measurement then set at 3.25X gear mesh frequency as shown in Table III. Very commonly, gearboxes may show low amplitudes at the fundamental gear mesh frequency (GMF), but may display very high amplitudes at 2X and/or 3X GMF. In addition, looseness is sometimes evidenced at one-half harmonics of gear mesh frequency, up to 2.5X GMF. Therefore, the maximum frequency is set at 3.25X GMF in order to allow for capture of gear mesh and accompanying sideband frequencies up thru 3X GMF.

Please note in Table III that spectra with 1600 to 3200 lines of resolution are recommended for these high frequency measurements. The reason for this is to allow 1X RPM sidebands to be displayed with good resolution around gear mesh frequency harmonics, not only for the high speed pinion, but also for the lower speed gear. Such high resolution spectra will also be recommended for Case D (when the tooth count is unknown).

A complete example illustrating specification of spectral alarm bands for a 2-stage speed increaser gearbox driving a compressor is given in Figure 11. Note the setups for both the lower frequency measurements (i.e., positions 3HI Axial and 3HI Horizontal) in Figure 11.

Please note the caution under Case C to keep in mind that a requirement to set F_{MAX} at 3.25X GMF may cause one to specify a maximum frequency that is not necessarily greater than the transducer frequency specifications, but can easily approach the natural frequency of the transducer mounting itself, causing errors in amplitude measurements. That is, when a transducer is mounted on a machine, it just creates another "spring/mass" in the system. The natural frequency of this spring/mass depends on how the transducer is mounted on the machine (stud, magnet, hand-held or extension probe). Stud mounting provides the highest natural frequency and, therefore, allows the highest measurable frequency with little or no deviation in amplitude readout. If forcing frequencies (such as gear mesh frequencies) are present close to the mounting natural frequency, considerable amplification can occur causing error in the amplitude readout, but not in the frequency. On the other hand, if forcing frequencies are above the mounting natural frequency, they can result not only in deviation in amplitude readout, but can also cause phase error since this transducer/mount system will experience almost a 180° difference in phase when it passes through resonance.

However, if this is kept in mind by the user, he can still take data at fairly high frequencies, being aware that amplitude levels may not be absolute. In any case, if they are repeatable, they can at least be trended; and, since frequency information remains correct, will likely allow the user to detect potential problems. If they are not repeatable, he will have to try a different transducer or method of mounting the original transducer.

Referring again to Case C in Table III, each of the frequency ranges are specified in terms of GMF multiples (for example, Band 2 extends from .75X GMF to 1.25X GMF). Here again, band alarms are set in terms of overall alarm percentage.

Importantly, if the gearbox has more than one set of individual meshes, as in the case of a double or a triple reduction unit, each set of high frequency points will need to employ the gear mesh frequency that applies at that particular measurement point. For example, if point A were close to the input gear mesh having a 100,000 CPM GMF and point B was at the output near a second mesh with a 25,000 CPM GMF, the high frequency point A would use the 100,000 CPM when setting up its bands (setting F_{MAX} at 325,000 CPM), whereas point B would employ the 25,000 CPM GMF (setting its F_{MAX} at approximately 81,250 CPM). This will be further illustrated in a gearbox example to be given later (Figure 8).

Case D - Gearbox High Frequency Points with Unknown Number of Teeth:

Unfortunately, in most programs, the number of teeth in the great majority of gearboxes is unknown. In many cases, even the operating speeds of intermediate gears are unknown. However, in spite of this, one can set up effective spectral alarm bands which can be used until the true number of teeth are confirmed (however, when the tooth count is found, the spectral alarm bands should be respecified as per Case C). Referring to Case D in Table III, note that a maximum frequency of 200X shaft RPM will apply to each high frequency gearbox point. Note that the shaft speed at each particular measurement point will be used in specifying frequency ranges for each of the 6 bands. For example, if the input speed at point A was 1000 RPM and the output speed at point B was about 200 RPM, F_{MAX} at point A would be set at 1000 RPM X 200 (200,000 CPM) while that at point B would be set at 200 RPM X 200 (40,000 CPM).

In many plants, both the number of teeth and intermediate speeds are unknown in many multi-stage gearboxes. One approach to this problem of determining what the gear mesh frequencies are might be to acquire several sets of spectra on the gearbox and compare them to the "TYPICAL SPECTRUM" shown in Table I, Case E for "Gear Misalignment" (note that it shows both GMF and 2GMF, each of which are sidebanded at 1X RPM). If two or three harmonics of a high frequency fundamental are found (for example, fundamental at 40 to 60X RPM), it is possible that these are gear mesh frequencies, particularly if they each have 1X RPM sidebands. However, one must keep in mind that this same signature pattern could be caused by another problem (for example, rolling element bearing frequency harmonics at, say, 5X and 10X inner race frequency).

Therefore, Table III, Case D suggests another approach if the number of teeth and intermediate speeds are unknown. In these cases, one normally knows at least the gearbox ratio, and therefore, the input and output speeds. The note in Case D shows how to handle this case in which equal speed increment steps are assumed until one knows more about the intermediate shaft speeds. For example, if all you knew were the input speed, output speed and gear ratio, use the following formula as a start until you know more:

$$\text{Speed Increment Factor} = (\text{Gear Ratio})^{1/m}$$

Where m = number of separate gear meshes

For example, for a triple reduction gearbox with:

Input RPM = 3594, Assumed Speed Increment = ?
Output RPM = 230, Assumed Interm. #1 RPM = ?
Gear Ratio = 15.625, Assumed Interm. #2 RPM = ?

Gear Ratio = 15.625; and $1/m = 1/3 = .3333$
(3 meshes)

Thus, Speed Increment Factor = $(15.625)^{.3333} = 2.50$
Assumed Interm. #1 Speed = $3594/2.50 = 1438 \text{ RPM}$
Assumed Interm. #2 Speed = $1438/2.50 = 575 \text{ RPM}$

Again, when intermediate shaft speeds are confirmed, use these speeds in Case D. And, when the tooth count is confirmed, change spectral band setups immediately back to those specified in Case C of Table III.

Like Case C, the same caution is given on keeping in mind the high frequency limits of the transducer and its mounting. Often, this requires one to stud mount or temporarily epoxy the transducers for these high frequency measurements.

Case E - AC Induction Motor Electrical Rotor Bar Pass Frequency Point
(single point usually taken on outboard motor bearing):

The specific purpose of this measurement on each motor is to detect the presence of 1X and 2X rotor bar pass frequencies which often are accompanied by 2X line frequency (7200 CPM) sidebands, 1X RPM sidebands, and even pole pass frequency sidebands (see Table I). The rotor bar pass frequency (RBPF) is equal to the number of rotor bars times motor RPM. High amplitudes at rotor bar pass frequencies suggest rotor bar looseness and/or rotor eccentricity, particularly when these frequencies are accompanied by the 2X line and/or pole pass frequency sidebands.

Please note that this data is only taken at one point on each motor (normally on the outboard horizontal housing). Notice that the maximum frequency for this point (F_{MAX}) is fixed at 360,000 CPM. Also, note that Band 1 begins at 30,000 CPM and incrementally takes 55,000 CPM steps in each succeeding band up to 360,000 CPM in Band 6 (independent of operating speed for this point which applies to 900 to 3600 RPM motors). Here again, recall that the standard points also taken on this motor (specified using either Case A or B, depending on the bearing type) will evaluate unbalance, misalignment, etc. The number of rotor bars in most all motors is rarely known, but normally ranges between about 35 to 95. Therefore, the F_{MAX} of 360,000 CPM should almost always encompass both the first and second harmonic rotor bar pass frequencies, even on 2-pole, nominal 3600 CPM motors. 1600 line spectra are recommended here with 8 to 16 spectrum averages. Since F_{MAX} is so high, even 16 averages of 1600 line spectra should require only about 7 to 10 seconds total. However, due to the high frequency, this might require permanent placement of a disk using a thin layer of high frequency epoxy adhesive in order to provide a dependable measurement mounting. Use of a high strength, rare-earth magnet is recommended in order to provide a good transducer mounting for this important electrical measurement.

Case F - AC Induction Motor Electrical Measurement Point
(single point usually taken on inboard motor bearing):

The whole purpose of this single point on each motor is to (1) attempt to separate mechanical and electrical vibration frequencies, particularly in the area of 1X RPM, line frequency (3600 CPM or 60 Hz), and 2X line frequency (7200 CPM or 120 Hz); and (2) to detect the possible presence of pole pass frequency sidebands around running speed harmonics. Very often in predictive maintenance programs, the spectrum will show high vibration at a so-called frequency of 7200 CPM which might suggest electrical problems. However, unless one has the required frequency resolution to separate running speed harmonics from the electrical synchronous frequencies, he cannot truly detect the presence of either a mechanical or an electrical problem, its severity, and certainly not its cause (variable air gap, stator problems, etc.). This is due to the fact that with an F_{MAX} of 50X RPM, he cannot separate, for example, the 3580 RPM operating speed peak from the 3600 CPM line frequency.

Therefore, if one uses 3200 FFT lines of resolution and a 12,000 CPM F_{MAX} , he will likely be able to separate most of these mechanical and electrical frequencies, depending on the motor RPM. Note that 400 FFT lines with a 12,000 CPM F_{MAX} will result in a 30 CPM frequency resolution which means peaks must be at least 90 CPM apart to show two separate frequencies (2X frequency resolution X 1.5 Hanning Noise Factor). For example, if the speed of a 2-pole motor is approximately 3550 RPM, one would be able to separate running speed from 3600 CPM line frequency and 2X RPM (7100 CPM) from 2X line frequency (7200 CPM) using only 400 lines and a 12,000 CPM F_{MAX} . However, if the motor speed were higher in the range of 3590 RPM (10 CPM slip frequency X 2 poles = 20 CPM pole pass frequency), it will require 3200 lines of resolution to display both the running speed harmonics and pole pass sidebands (3200 lines with a span of 12,000 CPM will give a frequency resolution of 3.75 CPM and a bandwidth of 5.625 CPM allowing the analyst to see each set of frequencies). Due to the high resolution of 3200 lines and rather

small frequency span of 12,000 CPM, this will require 16 seconds for the first average. Therefore, it is recommended that 50% overlap processing be used taking 2 averages for this measurement (which will result in 16 seconds for the first average and 8 seconds for the second average, or a total of 24 seconds). However, the end result of this measurement is critical. It alone will allow the analyst to separate mechanical and electrical problems, plus detect potentially serious cracked or broken rotor bars (which he cannot even detect using 400 lines and an F_{MAX} of 50X RPM or so). It will also allow detection of subsynchronous belt defect frequencies on belt-driven machinery.

Case G - Centrifugal Compressors, Blowers and Pumps

Driven components require a different setup of spectral band alarms than those specified for the Driving components which are covered in Cases A and B (i.e., motors, turbines, gearboxes, etc.). This section will cover various centrifugal machine types including pumps, blowers and compressors. For example, the primary purpose for building special bands for centrifugal machines outfitted with rolling element bearings is to attempt to separate the blade pass frequency band from the bearing defect frequency band. The problem here is that amplitudes which would be acceptable at blade pass frequencies (BPF) would normally be excessive for a bearing defect frequency. If both the blade pass frequency and bearing frequencies coexist within the same band, it would be impossible to separate the alarm levels for these two unique sources. This will be discussed in following sections below.

Note that Types 1 and 2 cover measurements on centrifugal machines outfitted with rolling element bearings while Types 3 and 4 cover such machines with sleeve (or journal) bearings. Types 1 and 3 assume the number of impeller blades (or vanes) is known whereas Types 2 and 4 assume the analyst does not yet know the number of blades (when the number of blades is confirmed, replace the setups for Types 2 or 4 immediately with Type 1 or 3 (depending on whether the measurement is on a rolling element or on a sleeve bearing). Spectral alarm band setups are much more meaningful and effective when the number of impeller blades is confirmed.

Type 1 - Driven Centrifugal Component with Known Number of Vanes (or Blades) and Rolling Element Bearings:

Type 1 will cover driven centrifugal machines outfitted with rolling element bearings where the number of vanes (or blades) in a pump, fan or compressor is known. In these cases, it will be possible to set up a separate band to capture blade pass frequency (BPF), allowing a higher alarm for it than that for the bands containing bearing defect frequencies (BPFI, BPFO, etc.). This procedure is illustrated in Table III under Type 1 of Case G. Band 4 will include the fundamental blade pass frequency (BPF) as well as 1X RPM sidebands above and below BPF. This band will have an alarm level of 60% of the overall alarm. On the other hand, bearing frequency Bands 3 and 5 on either side of Band 4 will have much lower alarms as seen in Type 1 of Case G. Notice that Band 5 will likely capture not only lower harmonic bearing frequencies, but also harmonics of blade pass frequencies which might relate to flow pulsation problems.

Type 2A - Centrifugal Pumps with Unknown Number of Vanes and Rolling Element Bearings:

Type 2A covers pumps outfitted with rolling element bearings when the number of impeller vanes is not known. In this case, the frequency limits for the probable BPF in Band 4 are set to capture what should be blade pass frequency for roughly 60% to 80% of centrifugal pumps which often have 4 to 6 vanes. Of course, if this is not the case, Band 4 can be adjusted. In any case, when the number of vanes is found, replace the spectral alarm bands shown here with those given in Type 1. Notice in Type 2A that the Band 4 alarm is set at 60% of the overall alarm as in the case of Type 1. Here, the intention is to ensure that if fundamental bearing frequencies do occur within this band, amplitudes will not be allowed to become highly excessive before a potential bearing problem is detected. Fortunately, even if fundamental bearing frequencies do coexist with blade pass within Band 4, worn bearings typically will generate several harmonics of bearing defect frequencies exceeding the higher frequency of Band 4 where alarm levels will be much lower (see

Bands 5 and 6 in Type 2A of Case G). Figure 2 shows where measurement locations should be established on the bearing caps of a centrifugal pump. Take care not to mistake a seal or packing gland for a bearing.

Type 2B - Centrifugal Blowers & Compressors with Unknown Number of Blades and Rolling Element Bearings:

Spectral band setups in Type 2B show that experience has proven the Blade Pass Frequency amplitudes for most blowers and centrifugal compressors are typically lower than BPF amplitudes on a centrifugal pump (note the Band 4 alarm of 40% of the Overall alarm not to exceed .100 in/sec for these centrifugal machines versus levels of 60% of the overall not to exceed .185 in/sec for pumps). In addition, there are typically more blades on blower and compressor impellers than on pump impellers.

Thus, the "assumed" BPF for Type 2B is expected to be in the vicinity of 8X to 12X RPM. Of course, once the actual number of blades is ascertained, the analyst is instructed to immediately replace this Case 2B setup with that shown in Type 1 since a much better set of alarm bands will be employed in those cases where the BPF is confirmed. Figure 3 shows optimum locations for measurements on the bearing caps of centrifugal air compressors.

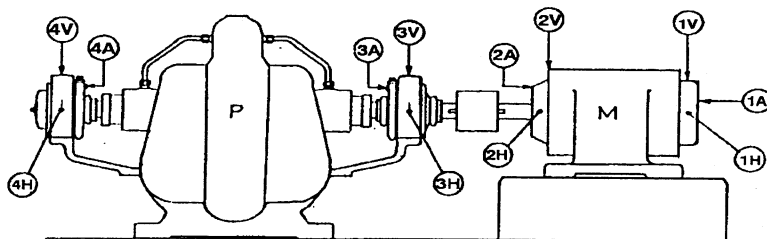


FIGURE 2 MEASUREMENT LOCATIONS ON CENTRIFUGAL PUMPS

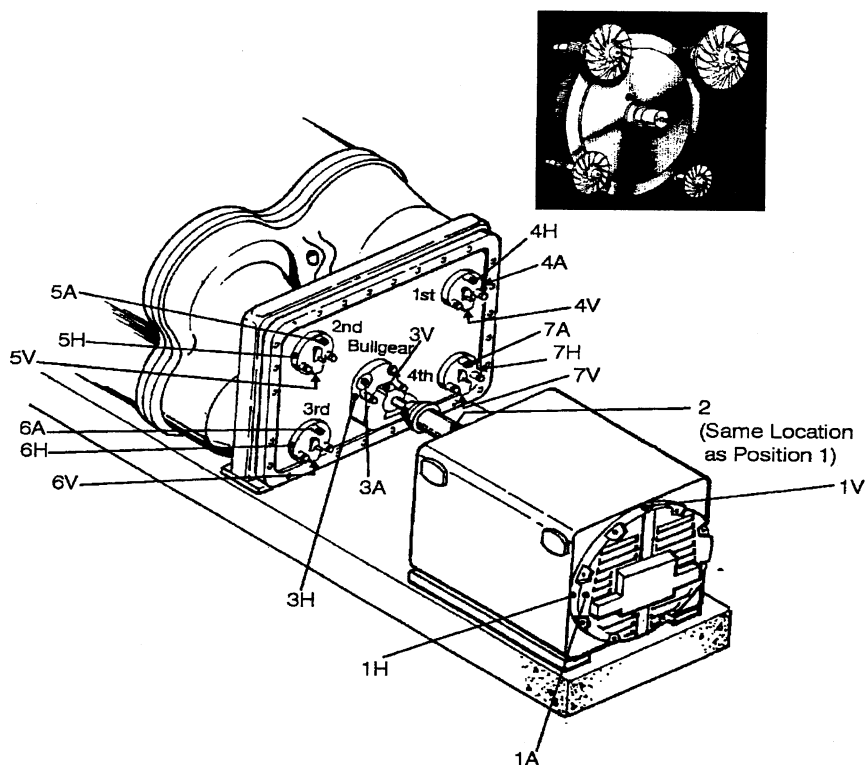


FIGURE 3 REQUIRED MEASUREMENT LOCATIONS ON CENTRIFUGAL AIR COMPRESSORS

Type 3 - Driven Centrifugal Component with Known Number of Vanes (or Blades) and Sleeve Bearings:

Types 3 and 4 differ from Types 1 and 2 in that these centrifugal machines are outfitted with sleeve (or journal) bearings rather than rolling element bearings. In this case, the maximum frequency will be set at only 20X RPM or 1.2X Blade Pass (whichever is greater), versus 50 to 120X RPM in the case of a unit outfitted with rolling element bearings. Here, Band 5 will capture the blade pass frequency (BPF) as well as 1X RPM sidebands above and below BPF. Band 5 will have an alarm set at 70% of the overall, versus only 30% of the overall alarm in the case of Bands 4 and 6 to its left and right. Here again, note that driven components require different spectral band alarm

Case H - DC Motors: Full-Wave and Half-Wave SCR Controlled Rectifier

DC motors controlled by silicon controlled rectifiers (SCR's) require special measurements in order to detect problems not only within the motors, but also those within the controls serving them (see "DC Motors and Controls" section of the "Diagnostic Chart" in Table I). The spectral setup parameters required to detect these faults are shown in Types 1, 2 and 3 of Section H in Table III. Each of these measurements require fine frequency resolution spectra in order to detect the faults within these machines and controls (3200 - 6400 lines). Figure 4 shows the general construction of a DC motor and where measurements are taken.

It should be noted that the spectral band setups shown in Case H assume a line frequency (F_L) of 60 Hz (3600 CPM). If the line frequency differs from 60 Hz, the frequency span of each band should likewise be adjusted. For example, if served by a line frequency of 50 Hz (3000 CPM), the analyst should center 600 CPM wide bands around 3000 CPM and its harmonics (i.e., 2700-3300 CPM in Band 1 for Types 1 and 2).

Note that Types 1, 2 and 3 are special measurements specifically used to detect electrical problems within DC motors and controls. These measurements should typically be taken only in the horizontal direction as per Case H of Table III. Standard spectral setups like those shown in Cases A and B should be taken in horizontal, vertical, and axial directions on both the outboard and inboard bearings of each DC motor in order to detect mechanical problems such as unbalance, misalignment, looseness, bearing problems, etc.

Type 1 - Full-Wave Rectified Motors (Measure on Commutator-End Bearing in the Horizontal Direction):

The purpose of this special measurement taken on the commutator end of full-wave rectified motors is to detect problems with armature windings, commutators, SCR's, firing cards, comparator cards, fuses, etc. In order to detect these problems, one must use 6400 lines and 2 averages with 50% overlap processing. This will allow the analyst not only to detect 1X RPM sidebands around the SCR firing frequency, but also Δ RPM sidebands which would likely be present if a DC motor has comparator card problems causing its speed to fluctuate. Note that the SCR firing frequency for a full-wave rectified motor is 21,600 CPM (360 Hz) assuming a line frequency (F_L) of 60 Hz. Looking at the spectral band setup shown in Type 1, note that amplitudes of only .02 in/sec are specified for Bands 1 thru 5. This is due to the fact that the first 5 line frequency harmonics should not be present in a DC motor served by SCR controls. On the other hand, the SCR firing frequency ($6 F_L$) is expected to be seen just as 2X line frequency ($2 F_L$) in an AC motor or blade pass frequency (BPF) in a pump. However, problems are indicated if the amplitude at SCR firing frequency exceeds the alarm level. Note that the alarm level of .08 in/sec shown in Band 6 (covering SCR firing frequency and 1X RPM sidebands) has been determined by sampling large quantities of DC motor spectra, statistically processing the amplitudes at this frequency and determining appropriate alarm levels which were noted when certain faults were found present with either DC motors or the controls serving them.

Type 2 - Half-Wave Rectified Motors (Measure on Commutator-End Bearing in the Horizontal Direction):

This measurement is very similar to that shown for Type 1 with the exception that this spectral setup covers half-wave rectified motors where the SCR firing frequency is 10,800 CPM (180 Hz) assuming a line frequency (F_L) of 60 Hz. Looking at the spectral band setups for Type 2 in Table III, note that alarm levels of only .02 in/sec are specified for 1X line and 2X line frequencies since these frequencies should not be present within a SCR controlled DC motor; similarly, levels of only .015 in/sec are specified at 4X line and 5X line frequency for these half-wave rectified motors. Note that the alarm specified for the SCR firing frequency (3X line frequency for half-wave rectified motors) is again .08 in/sec as determined by comprehensive statistical compilations on half-wave rectified motors.

Type 3 - Special Measurement Point Required For Detection of Electrically-Induced Fluting on Full-Wave or Half-Wave Rectified DC Motors:

The primary purpose of this special measurement taken on both the outboard and inboard bearings of a DC motor is to detect possible electrically-induced fluting which is caused by electrical current flow through bearings. The "Diagnostic Chart", Table I, shows that fluting is normally detected by a series of difference frequencies modulating a carrier frequency in the neighborhood of 100,000 to 150,000 CPM. These difference frequencies will most often occur at the outer race defect frequency (BPFO), but can occasionally appear at the inner race frequency (BPFI) as well. Note that if only standard measurements are taken on DC motors up to approximately 50X RPM, the presence of electrically-induced fluting will very likely not be detected. As the spectral criteria in Type 3 shows, a frequency range extending to 180,000 CPM (3000 Hz) should be specified, along with a frequency resolution of 3200 lines and at least 8 averages. Typically, this data should be taken in the horizontal direction on both the outboard and inboard bearings since such fluting can occur on either or both bearings. However, if a horizontal measurement would be distant from the bearing, whereas an axial measurement is either on or much closer to the bearing, the measurement should be taken in the axial direction due to the rather high frequency range required. In addition, the analyst should ensure the transducer is well mounted due to the rather high maximum frequency of 180,000 CPM. Alarm levels for each of the 6 bands are shown in Type 3 of Case H in Table III.

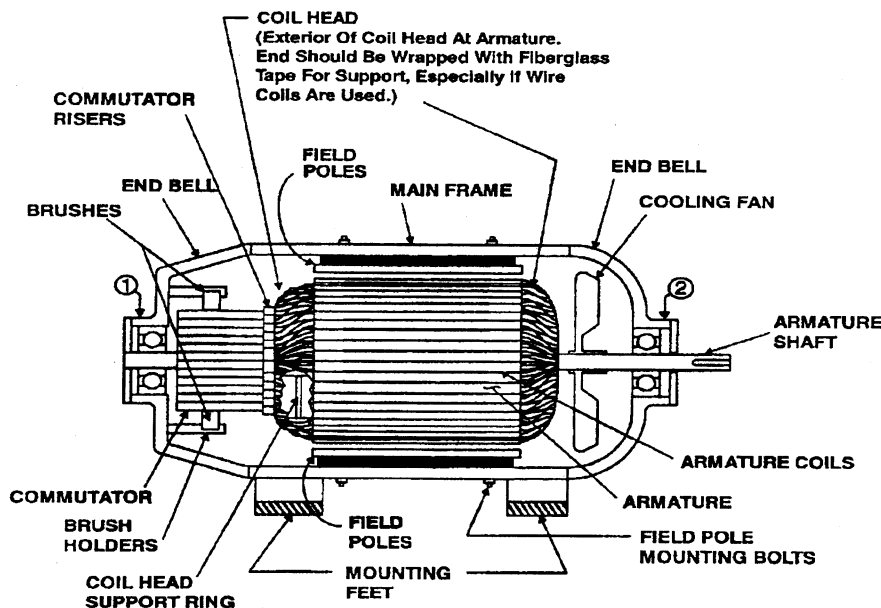


FIGURE 4. GENERAL DC MOTOR CONSTRUCTION (Ref. 41)

Case I - Machine Tool Spindles and Multi-Gear Heads (Ref. 40)

Machine tools are used to produce a wide variety of products. Machine tools are constructed of high quality components to maintain rigorous tolerances and to withstand significant stresses imposed upon them during machining processes. These machines typically have some of the lower vibration amplitudes in comparison with most all other machine types.

Table II includes overall vibration levels anticipated for machine tools. Note particularly the "Alarm 1" and "Alarm 2" levels specified for machine tool spindles. Three different grades are specified in Table II with the following "Alarm 1" amplitudes:

- | | | |
|----|-------------------------------|-------------|
| a. | Roughing Operations | .100 in/sec |
| b. | Machine Finishing Operations | .060 in/sec |
| c. | Critical Finishing Operations | .040 in/sec |

Figures 5 thru 9 include a variety of machine tool types (these drawings are extracted from Ref. 40). These include each of the following machine tool types:

<u>SPINDLE TYPE</u>	<u>FIG. NO.</u>
Precision Box Spindle	Fig. 5
Precision Cartridge-Type Spindle	Fig. 6
Precision Multi-Gear Head Spindle	Fig. 7
Precision Multi-Gear Head Gearbox	Fig. 7
Precision Grinding Wheel Head	Fig. 8
Precision Integral-Motor Spindle	Fig. 9

Following below are brief descriptions of each of the machine tools listed above, along with drawings of each spindle type covered:

Precision Box Spindles (Fig. 5) - Driven by belts or couplings and include foot-mounted or block design types. Precision Box Spindles may be used in, but not limited to, drilling, reaming, chamfering, spot fishing, countersinking, boring, milling, and single-wheel grinding operations. Figure 5 shows an example of a Precision Box Spindle.

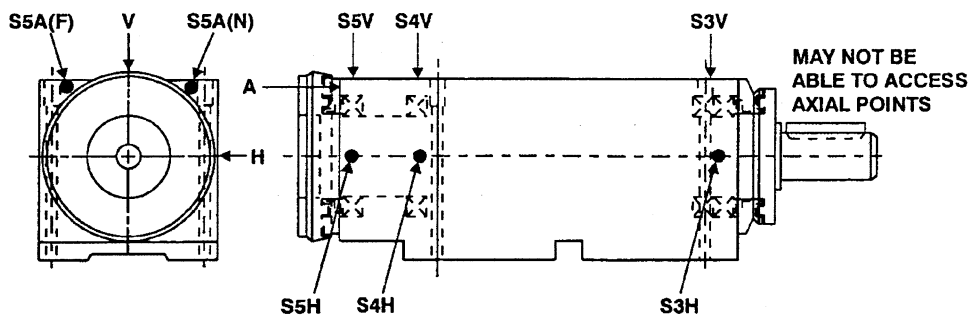


FIGURE 5
EXAMPLE OF CONSECUTIVE MEASUREMENT POINT LAYOUT
ON PRECISION BOX SPINDLES
(REF. Setco 4300B Series Spindles, Catalog #181D, p. 9)

Precision Cartridge-Type Spindles (Fig. 6) - Driven by belts or couplings and include plain or flange design types. Precision Cartridge-Type Spindles may be used in, but not limited to, drilling, boring, milling, turning, and internal deep-hole grinding operations. Figure 6 shows an example Precision Cartridge-Type Spindle. Note that in cases where the outboard bearings are the only accessible bearings, horizontal, vertical and axial measurements should be taken on this outboard bearing location only (per spindle). See Figure 6.

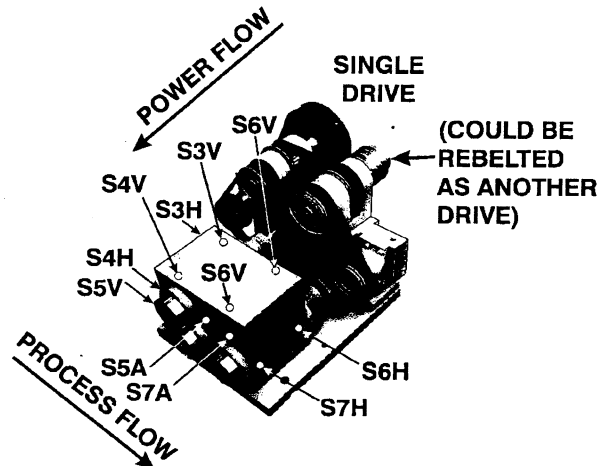


FIGURE 6
EXAMPLE OF CONSECUTIVE MEASUREMENT POINT LAYOUT ON
PRECISION CARTRIDGE-TYPE SPINDLES
 (REF. Setco Cluster Spindles, Catalog #181D, Photo #3060, p.39)

Precision Multi-Gear Heads (Fig. 7) - Precision Multi-Gear Heads are typically driven by belts or couplings. They include parallel-axis spindles and worm-gear spindles in integral housings or external housings such as the bolt-on, sub-plate or slide-assembly design types. They may be used in, but not limited to, cluster drilling, reaming, chamfering, spot-facing, countersinking and milling operations. It is recommended that measurement locations be numbered in the direction of power flow as well as in the direction of process flow as shown in Figure 7.

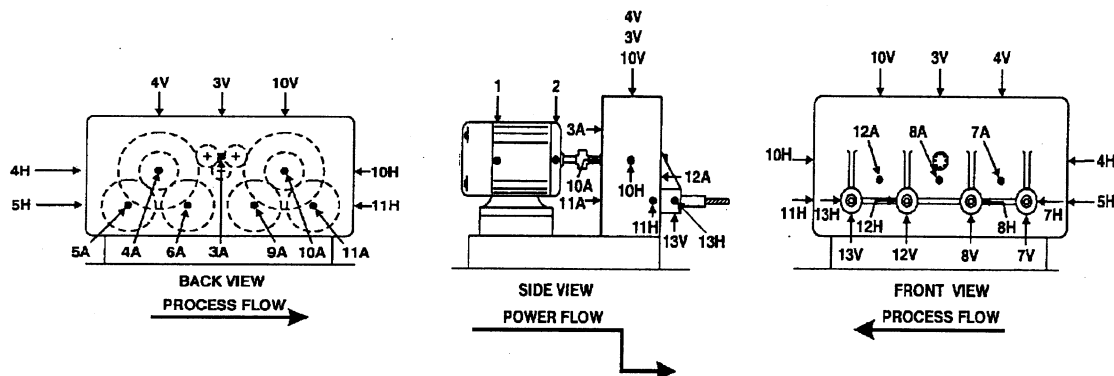


Figure 7.
EXAMPLE OF CONSECUTIVE MEASUREMENT
POINT LAYOUT ON PRECISION MULTI-GEAR HEADS

Precision Grinding Wheel Heads (Fig. 8) - Precision Grinding Wheel Heads are usually belt-driven and include the center and centerless design types. They may consist of, but are not limited to internal cylindrical and external cylindrical, surface, chucking as well as tool and cutter grinding operations, in horizontal or vertical orientations.

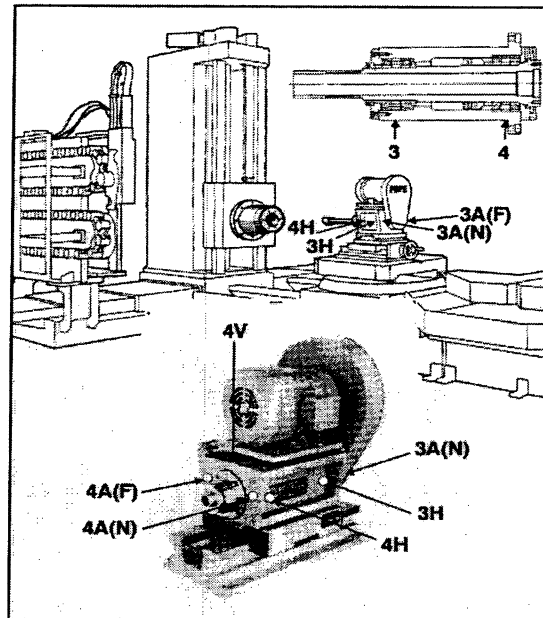


Figure 8.
EXAMPLE OF CONSECUTIVE MEASUREMENT
POINT LAYOUT ON PRECISION GRINDING WHEEL HEADS
 (REF. Pope Precision Spindles, SK-1418, Catalog No. 400, p.4)

Precision Integral-Motor Spindles (Fig. 9) - An example of a Precision Integral Motor-Spindle is shown in Figure 9. These units include the servo-design types. They may be used in, but not limited to drilling, reaming, chamfering, spotfacing, countersinking, boring, milling and internal deep-hole grinding operations. It is important that a cross-section view like that shown in Figure 9 is obtained for these spindles to ascertain the locations of bearings and to pinpoint good measurement locations relative to these bearings as shown in Figure 9.

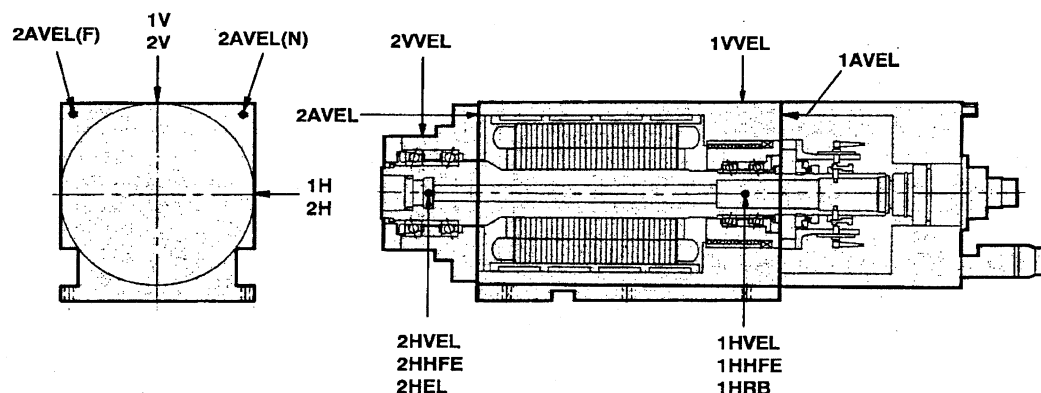


Figure 9.
EXAMPLE OF CONSECUTIVE MEASUREMENT POINT LAYOUT
ON PRECISION INTEGRAL MOTOR SPINDLES
 (REF. Setco Closed Loop Motorized Spindles, 230 Series, Catalog 995, p.10)

Vibration Alarms for Machine Tool Spindles:

Table II includes overall vibration alarm levels for machine tools. Note that the analyst must first determine the precision grade of the machine tool spindle in order to properly specify the overall alarm values for the spindle for which spectral band alarms are being specified ("Roughing Operations", "Machine Finishing", or "Critical Finishing" types). In addition, Section I on page 5 of Table III shows spectral band alarm measurement setups for such machine tool spindles based on the spindle type chosen. Note that alarm amplitudes for each of the six velocity bands are given in Table III and are expressed as a percentage of this overall alarm. Also, note that these alarm levels are broken down between bearing types (angular contact versus tapered roller bearings). For example, referring to Table III, note that alarm levels are slightly higher for tapered roller bearings than for angular contact bearings. When problems occur on bearings supporting these spindles, bearing fault frequencies and/or bearing natural frequencies are expected to be generated, particularly in Bands 5 and 6 of Case I. Higher alarm levels are specified for Bands 5 and 6 for tapered roller versus angular contact bearing since higher amplitude defect frequencies are typically generated by tapered roller bearings than those by angular contact bearings.

Table IIIB provides specific alarm levels at bearing fault frequencies (or harmonics) for each of the six machine tool types. These are separately specified for angular contact versus cylindrical and tapered roller bearings. For example, in the case of precision box spindles, Table IIIB shows that alarm levels of .005 in/sec should be applied for spindles supported by angular contact bearings versus .0075 in/sec for those supported by tapered roller bearings. Note that these specific alarm amplitudes apply at any bearing fault frequency or fault frequency harmonic (multiple) on any of the machine tool spindle types.

Table IIIC is also provided specifying maximum overall acceleration amplitudes for machine tool spindles (peak g's). These acceleration specifications all range between 600 and 600,000 CPM (10 - 10,000 Hz). Table IIIC specifies different acceleration alarm levels for angular contact versus tapered roller bearings; and in the case of tapered rollers, specifies higher acceleration levels for those operating above 1000 RPM (1.25 g) compared with those operating at or below 1000 RPM (0.75 g). This is due to the fact that higher acceleration amplitudes are generated by higher speed spindles supported by tapered roller bearings.

A note accompanying Table IIIC reminds analysts to take great care in mounting the transducer since measurements are acquired up to a rather high frequency of 600,000 CPM (10,000 Hz). Also, the analyst must ensure that the transducer itself is capable of taking accurate measurements up to this fairly high frequency.

7.151 EXAMPLES - Specification of Spectral Alarm Bands for Sample Machines:

Please refer to the Sample Machines shown in Figures 2 and 3, and then to the sample spectral alarm band tables which have been worked out for them. Note that the tables themselves are designed for direct input into the condition monitoring software once the alarm bands have been specified on them. Please note in particular how both the standard frequency and high frequency points are set up for the example with the 2-stage speed gear increaser for the gearbox shown in Figure 3.

TABLE II TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.

CRITERIA FOR OVERALL CONDITION RATING (PEAK OVERALL VELOCITY, IN/SEC)*

1. Assuming Machine Speed = 600 to 60,000 RPM.
2. Assuming Measurements by Accelerometer or Velocity Pickup securely mounted as Close as Possible to Bearing Housing.
3. Assuming Machine Is Not Mounted on Vibration Isolators (for Isolated Machinery - Set Alarm 30% to 50% Higher).
4. Set Motor Alarms the Same as that for the Particular Machine Type unless Otherwise Noted.
5. Consider Setting Alarms on Individual External Gearbox Position about 25% Higher than that for a particular Machine Type.

MACHINE TYPE	GOOD	FAIR	(WARNING)	(FAULT)
			ALARM 1	ALARM 2
COOLING TOWER DRIVES				
Long, Hollow Drive Shaft	0 - .375	.375 - .600	.600	.900
Close Coupled Belt Drive	0 - .275	.275 - .425	.425	.650
Close Coupled Direct Drive	0 - .200	.200 - .300	.300	.450
COMPRESSORS				
Reciprocating	0 - .325	.325 - .500	.500	.750
Rotary Screw	0 - .300	.300 - .450	.450	.650
Centrifugal With or W/O External Gearbox	0 - .200	.200 - .300	.300	.450
Centrifugal - Integral Gear (Axial Meas.)	0 - .200	.200 - .300	.300	.450
Centrifugal - Integral Gear (Radial Meas.)	0 - .150	.150 - .250	.250	.375
BLOWERS (FANS)				
Lobe-Type Rotary	0 - .300	.300 - .450	.450	.675
Belt-Driven Blowers	0 - .275	.275 - .425	.425	.650
General Direct Drive Fans (with Coupling)	0 - .250	.250 - .375	.375	.550
Primary Air Fans	0 - .250	.250 - .375	.375	.550
Vacuum Blowers	0 - .200	.200 - .300	.300	.450
Large Forced Draft Fans	0 - .200	.200 - .300	.300	.450
Large Induced Draft Fans	0 - .175	.175 - .275	.275	.400
Shaft-Mounted Integral Fan (Extended Motor Shaft)	0 - .175	.175 - .275	.275	.400
Vane-Axial Fans	0 - .150	.150 - .250	.250	.375
MOTOR/GENERATOR SETS				
Belt-Driven	0 - .275	.275 - .425	.425	.675
Direct Coupled	0 - .200	.200 - .300	.300	.450
CHILLERS				
Reciprocating	0 - .250	.250 - .400	.400	.600
Centrifugal (Open-Air) - Motor & Compressor Separate	0 - .200	.200 - .300	.300	.450
Centrifugal (Hermetic) - Motor & Impellers Inside	0 - .150	.150 - .225	.225	.350
LARGE TURBINE/GENERATORS				
3600 RPM Turbine/Generators	0 - .175	.175 - .275	.275	.400
1800 RPM Turbine/Generators	0 - .150	.150 - .225	.225	.350
CENTRIFUGAL PUMPS				
Vertical Pumps (12' - 20' Height)	0 - .325	.325 - .500	.500	.750
Vertical Pumps (8' - 12' Height)	0 - .275	.275 - .425	.425	.650
Vertical Pumps (5' - 8' Height)	0 - .225	.225 - .350	.350	.525
Vertical Pumps (0' - 5' Height)	0 - .200	.200 - .300	.300	.450
General Purpose Horizontal Pump - Direct Coupled	0 - .200	.200 - .300	.300	.450
Boiler Feed Pumps - Horizontal Orientation	0 - .200	.200 - .300	.300	.450
Piston Type Hydraulic Pumps - Horizontal Orientation (under load)	0 - .150	.150 - .250	.250	.375
MACHINE TOOLS				
Motor	0 - .100	.100 - .175	.175	.250
Gearbox Input	0 - .150	.150 - .225	.225	.350
Gearbox Output	0 - .090	.090 - .150	.150	.225
Spindles:				
a. Roughing Operations	0 - .065	.065 - .100	.100	.150
b. Machine Finishing	0 - .040	.040 - .060	.060	.090
c. Critical Finishing	0 - .025	.025 - .040	.040	.060

*NOTE: The "ALARM 1" and "ALARM 2" overall levels given above apply only to in-service machinery which has been operating for some time after initial installation and/or overhaul. They do not apply (and are not meant to serve as) Acceptance Criteria for either new or rebuilt machinery.

COPYRIGHT© 2000 - TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.

R-200002-PK

TABLE III RECOMMENDED SPECTRAL ALARM BANDS FOR VARIOUS MACHINE TYPES ("See Notes Below Table") Page 1 of 5

CASE	ITEM	BAND 1	BAND 2	BAND 3	BAND 4	BAND 5	BAND 6
A. GENERAL ROLLING ELEMENT BEARING COMPONENTS WITHOUT VANES - Set $F_{max} = 50X$ RPM (But, if Speed <1700 RPM or if Bearing is Tapered or Spherical Roller Type, See Notes b & c). (i.e., AC & DC Motors, Gearboxes) Gearboxes require second measurement with F_{max} and #Lines specified in Cases C and D.							
	BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARM	1% F_{max} 1.2 X RPM 90% OA ALARM	1.2 X RPM 2.2 X RPM 30% OA ALARM	2.2 X RPM 3.2 X RPM 25% OA ALARM	3.2 X RPM 12.2 X RPM 20% OA ALARM NTE .080 in/sec	12.2 X RPM 50% F_{max} 15% OA ALARM NTE .060 in/sec	50% F_{max} 100% F_{max} 10% OA ALARM NTE .050 in/sec
	DESCRIPTION OF BAND COVERAGE	(Subsynchronous thru 1X RPM)	(1.5 X - 2.0 X RPM)	(2.5 X - 3.0 X RPM)	(Fundamental Bearing Defect Frequencies)	(Lower Harmonic Bearing Frequencies)	(Higher Harmonic Bearing Freqs & Brg. Natl. Freqs.)
NOTES: a. Obtain Overall Alarm Velocity from Table II entitled "TECHNICAL ASSOCIATES CRITERIA FOR CONDITION RATING". Also " F_{max} " = Maximum Frequency for Spectrum (CPM or Hz). b. If bearing is a Tapered Roller Bearing or Spherical Roller Bearing, set $F_{max} = 80$ X RPM. c. If Speed = 1400-1700 RPM, set $F_{max} = 80$ X RPM; if 1100-1399 RPM, set $F_{max} = 80$ X RPM; if 800-1099 RPM, set $F_{max} = 100$ X RPM; if 600-799 RPM, set $F_{max} = 120$ X RPM. (To Detect Bearing Natural Frequencies)							
B. GENERAL SLEEVE BEARING MACHINES WITHOUT VANES (i.e., AC & DC Motors, Gearboxes) - Set $F_{max} = 20X$ RPM. (For Gearboxes, also see Case C and D).							
	BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARM	0.2 X RPM 0.8 X RPM 20% OA ALARM NTE .060 in/sec	0.8 X RPM 1.8 X RPM 90% OA ALARM	1.8 X RPM 2.8 X RPM 30% OA ALARM	2.8 X RPM 3.8 X RPM 25% OA ALARM	3.8 X RPM 10.2 X RPM 20% OA ALARM	10.2 X RPM 100% F_{max} 15% OA ALARM (10.5 X RPM - F_{max})
	DESCRIPTION OF BAND COVERAGE	(Subsynchronous Band)	(1 X - 1.5 X RPM)	(2 X - 2.5 X RPM)	(3 X - 3.5 X RPM)	(4 X - 10 X RPM)	
Gearboxes require second measurement with F_{max} and #Lines specified in Cases C and D.							
C. GEARBOX SPECIAL HIGH FREQUENCY POINTS WITH KNOWN NUMBER OF TEETH - Set $F_{max} = 3.25X$ Gear Mesh Frequency (GMF) - (However, see Case C Note Below)							
	BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARM	0.25 X GMF 0.75 X GMF 25% OA ALARM	0.75 X GMF 1.25 X GMF 70% OA ALARM	1.25 X GMF 1.75 X GMF 25% OA ALARM	1.75 X GMF 2.25 X GMF 50% OA ALARM	2.25 X GMF 2.75 X GMF 25% OA ALARM	2.75 X GMF 100% F_{max} 40% OA ALARM
	DESCRIPTION OF BAND COVERAGE	(0.5 X GMF & Sidebands)	(GMF & Sidebands)	(1.5 X GMF & Sidebands)	(2 X GMF & Sidebands)	(2.5 X GMF & Sidebands)	(3 X GMF & Sidebands)
NOTE: On Gearboxes, specify these high frequency points checking up to 3.25 X GMF as well as points with normal frequency ranges (as in Case A or B above) on these same points (i.e., 2 separate measurements on same point with 2 different frequency ranges). Specify $F_{max} = 3.25$ X GMF and locate points close to mesh locations of each set of gears using the specific gear mesh frequency at that particular location. However, remember that amplitudes at high frequencies approaching or exceeding the natural frequency of accelerometer mounting may be questionable (of transducer stud, magnet or probe mounting). In many cases, transducer may require stud or adhesive mounting and measurement of acceleration due to high F_{max} . Use 1800-3200 FFT Lines in order to allow detection of 1 X RPM Sidebands at each meshing gear.							
D. GEARBOX HIGH FREQUENCY POINTS WITH UNKNOWN NUMBER OF TEETH - Set $F_{max} = 200X$ Shaft RPM at that particular location (However, see Notes a thru d)							
	BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARM	20 X RPM 50 X RPM 60% OA ALARM	50 X RPM 80 X RPM 60% OA ALARM	80 X RPM 110 X RPM 50% OA ALARM	110 X RPM 140 X RPM 50% OA ALARM	140 X RPM 170 X RPM 40% OA ALARM	170 X RPM 100% F_{max} 40% OA ALARM (170X RPM - F_{max})
	DESCRIPTION OF BAND COVERAGE	(20X - 50X RPM)	(50X - 80X RPM)	(80X - 110X RPM)	(110X - 140X RPM)	(140X - 170X RPM)	
NOTE: a. Specify both high frequency as well as standard frequency measurement points at each location as in Case C. In many cases, transducer may require stud or adhesive mounting and measurement of acceleration due to high F_{max} . b. Use 3200 FFT Lines to detect 1X RPM Sidebands of each meshing gear. c. RPM = Shaft speed at the particular measurement point for which you are specifying spectral bands. If only input and output gearbox speeds are known in a multi-stage gearbox, assume an equal speed increment at each mesh (for example, for a 16:1 ratio, double reduction gearbox, assume approx. a 4.0 speed ratio at each of the mesh points as a starting point - refine when actual speeds are known).							

***NOTES:** 1. These Spectral Alarm Band Specs apply to "ALARM 1" for standard process and utility machinery such as centrifugal pumps, blowers, motors, FD & ID fans, motor/generator sets, refrigeration chillers, centrifugal air compressors, vacuum pumps, boiler feed pumps, gearboxes, etc. "ALARM 2" (higher alarm band setting) should have alarm levels 50% greater than "ALARM 1".
2. These Specs do not apply to other machine types such as reciprocating, axial screw or lobe-type rotary compressors; diesel engines; gas turbines; large turbine/generators; exciters; rotary blowers; pulverizers, etc. Special Bands for these types normally have to be custom-designed for each of these particular machines or group of similar machines, often after complete vibration spectra have been captured one or more times on these machines.
3. "NTE" = Not to Exceed; "OA ALARM" = Overall Alarm Specified in Table I.
4. Frequency Resolution = 800 Lines unless otherwise instructed; if $F_{max} > 100X$ RPM, increase resolution to 1600 Lines unless otherwise instructed; Use 50% Overlap Averaging.

TABLE III

RECOMMENDED SPECTRAL ALARM BANDS FOR VARIOUS MACHINE TYPES (*See Notes Below Table)

Page 2 of 5

CASE	ITEM	BAND 1	BAND 2	BAND 3	BAND 4	BAND 5	BAND 6
E. AC INDUCTION MOTOR ELECTRICAL ROTOR BAR PASS FREQUENCY POINT (MEASURE ON MOTOR OUTBOARD BEARING HORIZONTAL) - Set F_{max} = 360,000 CPM (See Notes a thru d).							
	BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARM	30,000 CPM 85,000 CPM .060 in/sec.	85,000 CPM 140,000 CPM .055 in/sec.	140,000 CPM 195,000 CPM .050 in/sec.	195,000 CPM 250,000 CPM .040 in/sec.	250,000 CPM 305,000 CPM .030 in/sec.	305,000 CPM 360,000 CPM .030 in/sec.
	DESCRIPTION OF BAND COVERAGE	(30K - 85K)	(85K - 140K)	(140K - 195K)	(195K - 250K)	(250K - 305K)	(305K - 360K)
NOTES: a. Purpose of this Point on each motor is to detect possible 1X and 2X Rotor Bar Pass Frequencies (RBPFF) which may be accompanied by 2X line frequency sidebands (normally at 7200 CPM). b. Case E applies to motors ranging in speeds from nominal 900 to 3600 RPM. c. Ensure that transducer is well mounted on the motor housing due to the rather high F_{max} . Normally, it should be taken in the horizontal direction on the outboard motor bearing housing. d. Should use 1600 FFT Lines & 8 Averages to clearly identify possible Rotor Bar Pass Frequencies as well as 2X Line Frequency and/or 1X RPM Sidebands.							
F. AC INDUCTION MOTOR ELECTRICAL MEASUREMENT POINT ON MOTOR INBOARD BEARING HORIZONTAL - Set F_{max} = 24,000 CPM (For nominal 900 - 3600 RPM motors).							
	BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARM	1.8 X Line Freq. 2.2 X Line Freq. .07 in/sec	3.8 X Line Freq. 4.2 X Line Freq. .035 in/sec	NOT REQUIRED	NOT REQUIRED	NOT REQUIRED	NOT REQUIRED
	DESCRIPTION OF BAND COVERAGE	(2 F_{max} $\pm$ Pole Pass Freq. Sidebands)	(4 F_{max} $\pm$ Pole Pass Freq. Sidebands)				
NOTES: a. Purpose of this Point is to attempt to separate Mechanical and Electrical vibration peaks, particularly in vicinity of 2X Line Frequency and 4X Line Frequency. b. Should use 6400 FFT Lines and 2 Averages to clearly separate Mechanical & Electrical Frequencies and to detect Pole Pass Freq. Sidebands (F_r = Slip Freq. X #Poles). c. This measurement is Required on Each Motor to Detect Pole Pass Freq. Sidebands surrounding either 1X RPM Harmonics and/or 2X Line Frequency.							
G. CENTRIFUGAL COMPRESSORS, BLOWERS AND PUMPS							
TYPE 1. DRIVEN CENTRIFUGAL COMPONENT WITH KNOWN NUMBER OF VANES (OR BLADES) AND ROLLING ELEMENT BEARINGS - Set F_{max} = 50X RPM (See Notes a thru e).							
	BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARM	0.4 X RPM 1.2 X RPM 90% OA ALARM	1.2 X RPM 2.2 X RPM 30% OA ALARM	2.2 X RPM (BPF - 1.2 X RPM) 35% OA ALARM NTE .120 in/sec.	BPF - 1.2 X RPM BPF + 1.2 X RPM 50% F_{max} 35% OA ALARM NTE .120 in/sec.	BPF + 1.2 X RPM 50% F_{max} 20% OA ALARM NTE .070 in/sec.	50% F_{max} 100% F_{max} 20% OA ALARM NTE .070 in/sec.
	DESCRIPTION OF BAND COVERAGE	(Subsynchronous thru 1 X RPM)	(1.5 - 2.0 X RPM)	(2.5 X - Funda. Bearing Defect Frequencies)	(Lower Harmonic Bearing Freqs. & BPF Harmonics)	(Higher Harmonic Bearing Freqs. & Brg. Natl. Freqs.)	
NOTES: a. Specify separate set of spectral alarm bands for Driver (as shown in Case A or B for motor, gearbox, etc.) where BPF is not involved. b. "BPF" = Blade Pass Frequency = No. of Blades (or Vanes) X RPM. If bearing is a Tapered Roller Bearing or Spherical Roller Bearing, set F_{max} = 60X RPM unless speed < 1700 RPM (see note b in Type 1). c. If Speed = 1400-1700 RPM, set F_{max} = 60X RPM. If 1100-1399 RPM, set F_{max} = 80X RPM. If 800-1099 RPM, set F_{max} = 100 X RPM. If 600-799 RPM, set F_{max} = 120X RPM.							
TYPE 2A. CENTRIFUGAL PUMPS WITH UNKNOWN NUMBER OF VANES AND ROLLING ELEMENT BEARINGS - Set F_{max} = 50X RPM (See Notes a thru c).							
	BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARM	1% F_{max} 1.2 X RPM 90% OA ALARM	1.2 X RPM 2.2 X RPM 30% OA ALARM	2.2 X RPM 3.2 X RPM 25% OA ALARM	3.2 X RPM 6.8 X RPM 60% OA ALARM NTE .185 in/sec.	6.8 X RPM 50% F_{max} 30% OA ALARM NTE .120 in/sec.	50% F_{max} 100% F_{max} 20% OA ALARM NTE .070 in/sec.
	DESCRIPTION OF BAND COVERAGE	(Subsynchronous thru 1 X RPM)	(1.5 - 2.0 X RPM)	(2.5-3.0 X RPM)	(Possible BPF for Pumps)	(Lower Harmonic Bearing Freq. & BPF Harmonics)	(Higher Harmonic Bearing Freqs. & Brg. Natl. Freqs.)
NOTES: a. If bearing is a Tapered roller Bearing or Spherical Roller Bearing, set F_{max} = 60 X RPM unless Speed < 1700 RPM. b. If Speed = 1400-1700 RPM, set F_{max} = 80X RPM; If 1100-1399 RPM, set F_{max} = 80X RPM; If 800-1099 RPM, set F_{max} = 100 X RPM; If 600-799 RPM, set F_{max} = 120X RPM. c. When Number of Vanes is confirmed, replace these Spectral Alarm Bands IMMEDIATELY with setup shown in Case G, Type 1.							

*NOTE: 1. These Spectral Alarm Bands apply to "ALARM 1" for standard process and utility machinery such as centrifugal pumps, blowers, motors, FD & ID fans, motor/generator sets, refrigeration chillers, centrifugal air compressors, vacuum pumps, boiler feed pumps, gearboxes, etc. "ALARM 2" (higher alarm band setting) should have alarm levels 50% greater than "ALARM 1".
 2. These Specs do not apply to other machine types such as reciprocating, axial screw or lobe-type rotary compressors; diesel engines; gas turbines; large turbine/generators; exciters; rotary blowers; pulverizers, etc. Spectral Bands for these types normally have to be "custom-designed" for each of these particular machines or group of similar machines, often after complete vibration spectra have been captured one or more times on these machines.
 3. "NTE" = Not to Exceed. "OA ALARM" = Overall Alarm Specified in Table II.
 4. Frequency Resolution = 800 Lines unless otherwise instructed; if F_{max} > 100X RPM, increase resolution to 1600 lines unless otherwise instructed; Use 50% Overlap Averaging.

© COPYRIGHT 2000

TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.

R-200002-4PK

TABLE III

RECOMMENDED SPECTRAL ALARM BANDS FOR VARIOUS MACHINE TYPES (*See Notes Below Table)

Page 3 of 5

CASE	ITEM	BAND 1	BAND 2	BAND 3	BAND 4	BAND 5	BAND 6
TYPE 2B. CENTRIFUGAL BLOWERS & COMPRESSORS WITH UNKNOWN NUMBER OF BLADES AND ROLLING ELEMENT BEARINGS - Set F_{max} = 50X RPM (See Notes a thru c).							
BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARM	1% F_{max} 1.2 X RPM 90% OA ALARM	1.2 X RPM 3.2 X RPM 35% OA ALARM	3.2 X RPM 7.8 X RPM 30% OA ALARM	7.8 X RPM 12.8 X RPM 40% OA ALARM	12.8 X RPM 50% F_{max} 20% OA ALARM	50% F_{max} 100% F_{max} 15% OA ALARM	50% F_{max} 100% F_{max} 15% OA ALARM
DESCRIPTION OF BAND COVERAGE	(Subsynchronous thru 1X RPM)	(1.5 - 3.0X RPM)	(3.5 - 7.5 X RPM)	(Possible BPF for Fans & Compressors)	(Lower Harmonic Bearing Freq. & BPF Harmonics)	(Higher Harmonic Bearing Freq. & BPF Harmonics)	(Higher Harmonic Bearing Freq. & BPF Harmonics)
NOTES: a. If bearing is a Tapered Roller Bearing or Spherical Roller Bearing, Set F_{max} = 60 X RPM unless Speed < 1700 RPM. (see Note b). b. If Speed = 1400-1700 RPM, set F_{max} = 80X RPM; if 1100-1399 RPM, set F_{max} = 80X RPM; if 800-1099 RPM, set F_{max} = 100 X RPM ² ; if 600-799 RPM, set F_{max} = 120X RPM ² . c. When Number of Blades is confirmed, replace these Spectral Alarm Bands IMMEDIATELY with setup shown in Case G, Type 1.							
TYPE 3. DRIVEN CENTRIFUGAL COMPONENT WITH KNOWN NUMBER OF VANES (OR BLADES) AND WITH SLEEVE BEARINGS - Set F_{max} = 20X RPM or 1.2X BPF, Whichever is Greater.							
BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARM	0.3 X RPM ² 0.8 X RPM 20% OA ALARM	0.8 X RPM 1.8 X RPM 90% OA ALARM	1.8 X RPM 3.8 X RPM 40% OA ALARM	3.8 X RPM (BPF - 1.2 X RPM) 30% OA ALARM	(BPF - 1.2 X RPM) (BPF + 1.2 X RPM) 70% OA ALARM	(BPF - 1.2 X RPM) (BPF + 1.2 X RPM) 100% F_{max}	(BPF - 1.2 X RPM) (BPF + 1.2 X RPM) 100% F_{max}
DESCRIPTION OF BAND COVERAGE	(Subsynchronous Band)	(1X - 1.5X RPM)	(2X - 3.5 X RPM)	(4X - Lower RPM Harmonics)	(BPF ± 1X Sidebands)	(Higher RPM Harmonics and BPF Harmonics)	(Higher RPM Harmonics and BPF Harmonics)
NOTES: a. Specify separate set of spectral alarm bands for Driver (as shown in Case A or B for motor, gearbox, etc.) where BPF is not involved. b. "BPF" = Blade Pass Frequency = No. of Blades (or Vanes) X RPM. c. If Number of Vanes > 25, set Band 1 Lower Frequency = 0.4 X RPM.							
TYPE 4A. CENTRIFUGAL, PUMPS EQUIPPED WITH UNKNOWN NUMBER OF VANES AND WITH SLEEVE BEARINGS - Set F_{max} = 20X RPM (See Notes a and b).							
BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARM	0.3 X RPM 0.8 X RPM 20% OA ALARM	0.8 X RPM 1.8 X RPM 90% OA ALARM	1.8 X RPM 3.8 X RPM 40% OA ALARM	3.8 X RPM 7.2 X RPM 70% OA ALARM	7.2 X RPM 9.8 X RPM 25% OA ALARM	9.8 X RPM 100% F_{max} 35% OA ALARM	9.8 X RPM 100% F_{max} 35% OA ALARM
DESCRIPTION OF BAND COVERAGE	(Subsynchronous Band)	(1X - 1.5X RPM)	(2X - 3.5X RPM)	(Possible BPF for Pumps)	(8X - 9.5X RPM)	(10X RPM - F_{max})	(10X RPM - F_{max})
NOTES: a. Specify separate set of spectral alarm bands for Driver (as shown in Case A or B for motor, gearbox, etc.) where BPF is not involved. b. When Number of Vanes is found, replace these Spectral Alarm Bands IMMEDIATELY with setup shown in Case G, Type 3.							
TYPE 4B. CENTRIFUGAL BLOWERS & COMPRESSORS WITH UNKNOWN NUMBER OF BLADES AND WITH SLEEVE BEARINGS - Set F_{max} = 30X RPM (See Notes a and b).							
BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARM	0.3 X RPM 0.8 X RPM 20% OA ALARM	0.8 X RPM 1.8 X RPM 90% OA ALARM	1.8 X RPM 3.2 X RPM 40% OA ALARM	3.2 X RPM 7.2 X RPM 30% OA ALARM	7.2 X RPM 12.8 X RPM 40% OA ALARM	12.8 X RPM 100% F_{max} 20% OA ALARM	12.8 X RPM 100% F_{max} 20% OA ALARM
DESCRIPTION OF BAND COVERAGE	(Subsynchronous Band)	(1X - 1.5X RPM)	(2X - 3X RPM)	(3.5 - 7.5X RPM)	(Possible BPF for Fans and Compressors)	(13X RPM - F_{max})	(13X RPM - F_{max})
NOTES: a. Specify separate set of spectral alarm bands for Driver (as shown in Case A or B for motor, gearbox, etc.) where BPF is not involved. b. When Number of Blades is found, replace these Spectral Alarm Bands IMMEDIATELY with setup shown in Case G, Type 3.							

*NOTE: 1. These Spectral Alarm Band Specs apply to "ALARM 1" for standard process and utility machinery such as centrifugal pumps, blowers, motors, FD & ID fans, motor/generator sets, refrigeration chillers, centrifugal air compressors, vacuum pumps, boiler feed pumps, gearboxes, etc. "ALARM 2" (higher alarm band setting) should have alarm levels 50% greater than "ALARM 1".
2. These Specs do not apply to other machine types such as reciprocating, axial screw or lobe-type rotary compressors; diesel engines; gas turbines; large turbine/generators; exciters; rotary blowers; pulverizers, etc. Spectral Bands for these types normally are custom-designed for each of these particular machines or group of similar machines, often after complete vibration spectra have been captured one or more times on these machines.
3. "NTE" = Not To Exceed "OA ALARM". Overall Alarm Specified in Table II.
4. Frequency Resolution = 800 Lines unless otherwise instructed; if F_{max} > 100X RPM, increase resolution to 1600 lines unless otherwise instructed; Use 50% Overlap Averaging.

© COPYRIGHT 2000

TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.

R-200002-4PK

RECOMMENDED SPECTRAL ALARM BANDS FOR VARIOUS MACHINE TYPES (*See Notes Below Table)

TABLE III

CASE	ITEM	BAND 1	BAND 2	BAND 3	BAND 4	BAND 5	BAND 6
H. DC MOTORS: FULL-WAVE AND HALF-WAVE SCR CONTROLLED RECTIFIER (Special Measurements in Addition to Standard Spectral Setups in Cases A & B).							
TYPE 1. FULL-WAVE RECTIFIED MOTORS (MEASURE ON COMMUTATOR-END BEARING HORIZONTAL): Set $F_{max} = 24,000$ CPM (See Notes a thru c).							
	BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARM	3300 CPM 3900 CPM .02 in/sec	6900 CPM 7500 CPM .02 in/sec	10,500 CPM 11,100 CPM .02 in/sec	14,100 CPM 14,700 CPM .02 in/sec	17,700 CPM 18,300 CPM .02 in/sec	21,600 - 1.2 X RPM 21,600 + 1.2 X RPM .08 in/sec
	DESCRIPTION OF BAND COVERAGE	(1 X Line Freq., F_c)	(2 X Line Freq.)	(3 X Line Freq.)	(4 X Line Freq.)	(5 X Line Freq.)	(SCR Firing Freq. ± 1 X RPM Sidebands)
NOTES:							
a. Purpose of this Special Measurement Point is to Detect Problems with Armature Windings, Commutators, SCR's, Firing Cards, Comparator Cards, Fuses, etc.							
b. Must use 6400 Lines and 2 Averages to Detect Δ RPM and 1 X RPM Sidebands around the SCR Firing Frequency; Use 50% Overlap Averaging.							
c. Assumes Line Frequency = 60 Hz (3600 CPM); Adjust Frequency Span of Each Band if Line Freq. Differs from 60 Hz.							
TYPE 2. HALF-WAVE RECTIFIED MOTORS (MEASURE ON COMMUTATOR-END BEARING HORIZONTAL) - Set $F_{max} = 24,000$ CPM (See Notes a thru c).							
	BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARM	3300 CPM 3900 CPM .02 in/sec	6900 CPM 7500 CPM .02 in/sec	10,800 - 1.2 X RPM 10,800 + 1.2 X RPM .08 in/sec	14,100 CPM 14,700 CPM .015 in/sec	17,700 CPM 18,300 CPM .015 in/sec	21,600 - 1.2 X RPM 21,600 + 1.2 X RPM .04 in/sec
	DESCRIPTION OF BAND COVERAGE	(1 X Line Freq., F_c)	(2 X Line Freq.)	(SCR Firing Freq. ± 1 X RPM Sidebands)	(4 X Line Freq.)	(5 X Line Freq.)	(2 X SCR Firing Freq. ± 1 X RPM Sidebands)
NOTES:							
a. Purpose of this Special Measurement Point is to Detect Problems with Armature Windings, Commutators, SCR's, Firing Cards, Comparator Cards, Fuses, etc.							
b. Must use 6400 Lines and 2 Averages to Detect Δ RPM and 1 X RPM Sidebands around the SCR Firing Frequency; Use 50% Overlap Averaging.							
c. Assumes Line Frequency = 60 Hz (3600 CPM); Adjust Frequency Span of Each Band if Line Freq. Differs from 60 Hz.							
TYPE 3. SPECIAL POINT FOR ELECTRICALLY - INDUCED FLUTING ON FULL-WAVE OR HALF-WAVE RECTIFIED DC MOTORS - SET $F_{max} = 180,000$ CPM (See Notes a thru e).							
	BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARM	(SCR - 2.2 X RPM) (SCR + 2.2 X RPM) .08 in/sec	(2 X SCR - 2.2 X RPM) (2 X SCR + 2.2 X RPM) .04 in/sec	(3 X SCR - 2.2 X RPM) (3 X SCR + 2.2 X RPM) .025 in/sec	(4 X SCR - 2.2 X RPM) (4 X SCR + 2.2 X RPM) .020 in/sec	(5 X SCR - 2.2 X RPM) 120,000 CPM .015 in/sec	120,000 CPM 180,000 CPM .015 in/sec
	DESCRIPTION OF BAND COVERAGE	(SCR Firing Freq. ± 2 X RPM Sidebands)	(2 X SCR Firing Freq. ± 2 X RPM Sidebands)	(3 X SCR Firing Freq. ± 2 X RPM Sidebands)	(4 X SCR Firing Freq. ± 2 X RPM Sidebands)	(BPFO or BPFI Sidebands Around Carrier Freq.)	(BPFO or BPFI Sidebands Around Carrier Freq.)
NOTES:							
a. Primary Purpose of this Special Measurement is to Detect Fluting normally indicated by BPFO and/or BPFI Sidebands around Carrier Frequency which Ranges from 100,000 to 150,000 CPM.							
b. SCR Firing Freq. for Half-Wave Rectified DC Motor = 10,800 CPM (180 Hz); but 21,600 CPM (360 Hz) for Full-Wave Rectified DC Motor (with a line frequency F_c of 60Hz).							
c. Must use 3200 lines & 8 averages with F_{max} of 180,000 CPM in order to detect electrically induced fluting which is indicated by Spectrum Described in Note a.							
d. Should take this measurement on Both the Outboard and Inboard Motor Bearings since fluting can occur on either bearing. Use 50% Overlap Averaging.							
e. Take data in Horizontal Direction on Both bearings and ensure transducer is well mounted due to the rather high F_{max} .							

*NOTE: 1. These Spectral Alarm Band Specs apply to "ALARM 1" for standard process and utility machinery such as centrifugal pumps, blowers, motors, FD & ID fans, motor/generator sets, refrigeration chillers, centrifugal air compressors, vacuum pumps, boiler feed pumps, gearboxes, etc. "ALARM 2" (higher alarm band setting) should have alarm levels 50% greater than "ALARM 1".
 2. These Specs do not apply to other machine types such as reciprocating, axial screw or lobe-type rotary compressors; diesel engines; gas turbines; large turbine/generators; exciters; rotary blowers; pulverizers, etc. Spectral Bands for these types normally have to be "custom-designed" for each of these particular machines or group of similar machines, often after complete vibration spectra have been captured one or more times on these machines.
 3. "NTE" = Not to Exceed; "OA ALARM" = Overall Alarm Specified in Table II.
 4. Frequency Resolution = 800 Lines unless otherwise instructed; if $F_{max} > 100X$ RPM, increase resolution to 1600 Lines unless otherwise instructed; Use 50% Overlap Averaging.

TABLE III

RECOMMENDED SPECTRAL ALARM BANDS FOR VARIOUS MACHINE TYPES (*See Notes Below Table)

Page 5 of 5

CASE	ITEM	BAND 1	BAND 2	BAND 3	BAND 4	BAND 5	BAND 6
I. MACHINE TOOL SPINDLES (see Table II and Text Section I) - Set $F_{max} = 50X$ RPM (But, if speed < 1700 RPM or if Bearing is Tapered or Spherical Roller Type, See Notes a thru c).							
BAND LOWER FREQ. BAND HIGHER FREQ. BAND ALARMS:	Angular Contact Tapered Roller	0.3 X RPM 0.8 X RPM	0.8 X RPM 1.2 X RPM	1.2 X RPM 2.2 X RPM	2.2 X RPM 3.8 X RPM	3.8 X RPM 50% F_{max}	50% F_{max} 100% F_{max}
		30% OA ALARM 30% OA ALARM	60% OA ALARM 60% OA ALARM	40% OA ALARM 40% OA ALARM	40% OA ALARM 40% OA ALARM	20% OA ALARM 30% OA ALARM	10% OA ALARM 15% OA ALARM
DESCRIPTION OF BAND COVERAGE	(FTF & Subsynch.)	(1X RPM)	(1.5X - 2.0X RPM)	(3X RPM & BSF)	(BSF BPFO, BPFI & Harmonics)	(Bearing/Housing Natural Freqs.)	

NOTES: a. Obtain Overall Alarm Velocity from Table II entitled "TECHNICAL ASSOCIATES CRITERIA FOR CONDITION RATING". Also " F_{max} " = Maximum Frequency for Spectrum (CPM or Hz).
b. If bearing is a Tapered Roller Bearing or Spherical Roller Bearing, set $F_{max} = 80 X$ RPM.
c. If Speed = 1400-1700 RPM, set $F_{max} = 80 X$ RPM; if 1100-1399 RPM, set $F_{max} = 80 X$ RPM; if 800-1099 RPM, set $F_{max} = 100 X$ RPM; if 600-799 RPM, set $F_{max} = 120 X$ RPM.
d. See Table IIIB which specifies maximum allowable amplitudes for discrete

TABLE IIIB. MAXIMUM ALLOWABLE AMPLITUDE AT BPFO, BPFI, BSF OR ANY BEARING FREQUENCY HARMONIC (Multiple) ON MACHINE TOOL SPINDLES (See Note below)

SPINDLE TYPE	TEXT FIG. NO.	MAX. BEARING FREQ. AMPLITUDE	
		ANGULAR CONTACT	CYLIND./ TAPERED
Precision Box Spindle	Fig. 2	.0050 in/sec	.0075 in/sec
Precision Cartridge-Type Spindle	Fig. 3	.0050 in/sec	.0075 in/sec
Precision Multi-Gear Head Spindle	Fig. 4	.0050 in/sec	N/A
Precision Multi-Gear Head Gearbox	Fig. 4	.0075 in/sec	.0075 in/sec
Precision Grinding Wheel Head	Fig. 5	.0050 in/sec	.0075 in/sec
Precision Integral-Motor Spindle	Fig. 6	.0050 in/sec	.0050 in/sec

NOTE: Information in Case I is derived from Technical Associates' "Vibration Acceptance Criteria for New and Rebuilt Machinery" (Ref. 40)

TABLE IIIC. MAXIMUM BAND LIMITED OVERALL PEAK ACCELERATION FOR MACHINE TOOL SPINDLES (See Note below)

BEARING TYPE	SPINDLE SPEED	FREQUENCY BAND*	PEAK ACCEL.
Angular Contact	Any Speed	600 - 600,000 CPM	0.50g
Tapered Roller	< 1000 RPM	600 - 600,000 CPM	0.75g
Tapered Roller	> 1000 RPM	600 - 600,000 CPM	1.25g

*NOTE: Great care must be taken in using a proper transducer and in mounting the transducer since measurements are required up to these high frequencies.

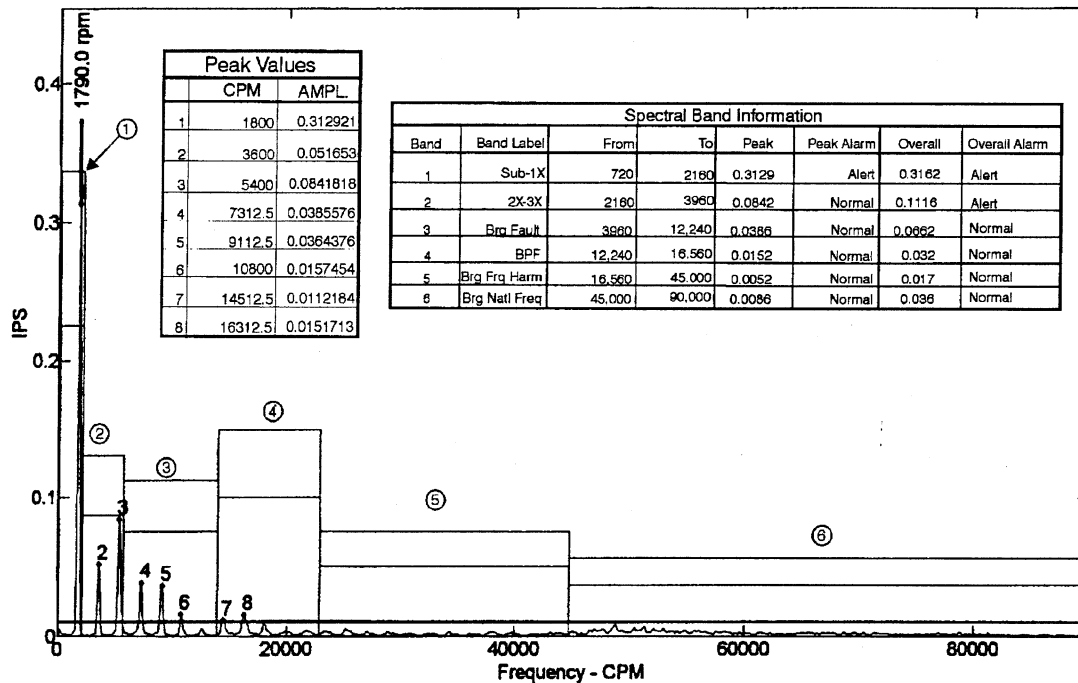
*NOTE: 1. These Spectral Alarm Band Specs apply to "ALARM 1" for standard process and utility machinery such as centrifugal pumps, blowers, motors, FD & ID fans, motor/generator sets, refrigeration chillers, centrifugal air compressors, vacuum pumps, boiler feed pumps, gearboxes, etc. "ALARM 2" (higher alarm band setting) should have alarm levels 50% greater than "ALARM 1".
2. These Specs do not apply to other machine types such as reciprocating, axial screw or lobe-type rotary compressors, diesel engines; gas turbines; large turbine/generators; exciters; rotary blowers; pulverizers, etc. Spectral Bands for these types normally have to be "custom-designed" for each of these particular machines or group of similar machines, often after complete vibration spectra have been captured one or more times on these machines.
3. "NTE" = Not to Exceed; "OA ALARM" = Overall Alarm Specified in Table II.
4. Frequency Resolution = 800 Lines unless otherwise instructed. If $F_{max} > 100X$ RPM, increase resolution to 1600 lines unless otherwise instructed. Use 50% Overlap Averaging.

© COPYRIGHT 2000

TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.

R-200002-4PK

FIGURE 10
SPECTRAL ALARM BAND SPECS ON A HORIZONTAL PUMP (8 Vanes)



EXAMPLE 1 TECHNICAL ASSOCIATES OF CHARLOTTE, P.C. (704) 333-9011

CLIENT NAME: XYZ COMPANY
MACHINE NAME: HORIZ. PUMP 2

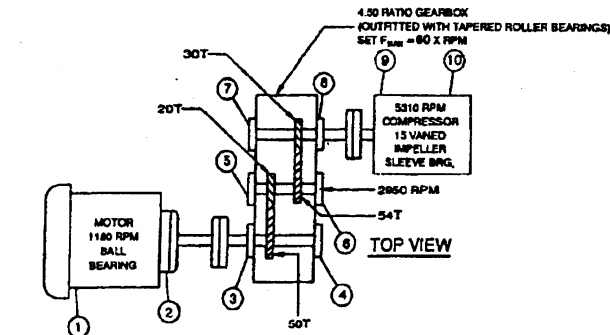
BY: JEB
DATE: 5/12/2000

SPECTRAL ALARM BAND COMPUTER ENTRY FORM

POSITION	UNITS	ALARM	RPM	BRGS.	F _{MAX}	#LINES	PARAMETER	BAND 1	BAND 2	BAND 3	BAND 4	BAND 5	BAND 6
1A, 1H, 1V 2A, 2H, 2V (Motor Points As Per Case A)	in/sec	.300	1780	Ball	90K	800	LOWER FREQ. HIGHER FREQ. BAND ALARM	900 2160 .270	2160 3960 .090	3960 5760 .075	5760 21,960 .060	21,960 45,000 .045	45,000 90,000 .030
1 BAR (Rotor Bar Pass Freq. Point From Case E)	in/sec	.300	1780	Ball	360K	1800	LOWER FREQ. HIGHER FREQ. BAND ALARM	30,000 85,000 .060	85,000 140,000 .055	140,000 195,000 .050	195,000 250,000 .040	250,000 305,000 .035	305,000 360,000 .030
2 EL (Motor Points As Per Case A)	in/sec	.300	1780	Ball	24K	6400	LOWER FREQ. HIGHER FREQ. BAND ALARM	6480 7920 .070	13,680 15,120 .035	N/A	N/A	N/A	N/A
1H, 2H, 3H, 4H	gSE	.700	1780	Ball	---	N/A	LOWER FREQ. HIGHER FREQ. BAND ALARM	NO ALARMS BANDS FOR HIGH FREQ. ENVEL. SPECTRA					
3A, 3H, 3V 4A, 4H, 4V (Pump Points As Per Case G, Type 1 with 8 Vanes)	in/sec	.300	1780	Ball	90K	800	LOWER FREQ. HIGHER FREQ. BAND ALARM	720 2160 .270	2160 3960 .090	3960 12,240 .105	12,240 16,560 .180	16,560 45,000 .105	45,000 90,000 .080

OTHER MACHINES WHICH WILL HAVE DUPLICATE SPECTRAL ALARM BANDS:			
HORIZ PUMP 3	HORIZ PUMP 12		
HORIZ PUMP 5	HORIZ PUMP 16		
HORIZ PUMP 7			
HORIZ PUMP 10			

FIGURE 11
EXAMPLE #2 - COMPRESSOR DRIVEN BY 2 - STAGE GEAR BOX



$$\begin{aligned}
 \text{GMF}_1 &= (50T)(1180 \text{ RPM}) = (20T)(2950 \text{ RPM}) & \text{GMF}_2 &= 54T \times 2950 \text{ RPM} = (30T)(5310 \text{ RPM}) \\
 \text{GMF}_1 &= 59,000 \text{ CPM} & \text{GMF}_2 &= 159,300 \text{ CPM} \\
 \text{POS. 3H } F_{\text{MAX}} &= \text{GMF} \times 3.25 & \text{POS. 6H } F_{\text{MAX}} &= 3.25 \times 159,300 \\
 &= 181,750 \text{ CPM} & &= 517,725 \text{ CPM} \\
 \text{USE POS 3H } F_{\text{MAX}} &= 200,000 \text{ CPM} & \text{USE POS 6H } F_{\text{MAX}} &= 640,000 \text{ CPM} \\
 & & &(\text{Must Measure Acceleration Here})
 \end{aligned}$$

EXAMPLE 2 **TECHNICAL ASSOCIATES OF CHARLOTTE, P.C. (704) 333-9011**

CLIENT NAME: XYZ COMPANY
 MACHINE NAME: COMPRESSOR2

BY: JEB
 DATE: 5/12/2000

SPECTRAL ALARM BAND COMPUTER ENTRY FORM

POSITION	UNITS	ALARM	RPM	BRGS.	F _{MAX}	#LINES	PARAMETER	BAND 1	BAND 2	BAND 3	BAND 4	BAND 5	BAND 6
1A, 1H, 1V 2A, 2H, 2V (From Case A)	in/sec	.300	1180	Ball	84K	800	LOWER FREQ. HIGHER FREQ. BAND ALARM	840 1440 .270	1440 2840 .090	2840 3840 .075	3840 14,840 .090	14,840 42,000 .045	42,000 84,000 .030
1 BAR (From Case E)	in/sec	.300	1180	Ball	360K	1800	LOWER FREQ. HIGHER FREQ. BAND ALARM	30,000 85,000 .090	85,000 140,000 .055	140,000 195,000 .050	195,000 250,000 .040	250,000 305,000 .035	305,000 360,000 .030
2 EL (From Case F)	in/sec	.300	1180	Ball	24K	8400	LOWER FREQ. HIGHER FREQ. BAND ALARM	8480 7920 .070	13,880 15,120 .035	N/A	N/A	N/A	N/A
1H, 2H, 3H, 4H	gSE	.500	1180	Ball	—	N/A	LOWER FREQ. HIGHER FREQ. BAND ALARM	NO ALARMS BANDS FOR HIGH FREQ. ENVEL. SPECTRA					
3A, 3H, 3V 4A, 4H, 4V (From Case A)	in/sec	.375	1180	Tapered Roller	84K	800	LOWER FREQ. HIGHER FREQ. BAND ALARM	840 1440 .340	1440 2840 .115	2840 3840 .095	3,840 14,840 .075	14,840 42,000 .055	42,000 84,000 .040
3H-I - A&H 5H-I - A&H (From Case C, Looking at GMF, (1st Mesh))	in/sec	.375	1180	Tapered Roller	200K	1800	LOWER FREQ. HIGHER FREQ. BAND ALARM	14,750 44,250 .340	44,250 73,750 .265	73,750 103,250 .095	103,250 132,750 .190	132,750 162,250 .065	162,250 200,000 .150
5A, 5H, 5V 6A, 6H, 6V (From Case A)	in/sec	.375	2950	Tapered Roller	180K	800	LOWER FREQ. HIGHER FREQ. BAND ALARM	1800 3540 .340	3540 6460 .115	6460 9440 .095	9440 36,000 .075	36,000 90,000 .055	90,000 180,000 .040
*6H-I - A&H 8H-I - A&H (From Case C, Looking at GMF, (2nd Mesh))	g Accel.	10.0g	2950	Tapered Roller	540K	1800	LOWER FREQ. HIGHER FREQ. BAND ALARM	36,800 118,500 2.0g	118,500 199,100 8.0g	199,100 278,800 3.0g	278,800 358,400 6.0g	358,400 438,100 3.0g	438,100 540,000 4.5g
5H, 6H	gSE	1.25	2950	Tapered Roller	—	N/A	LOWER FREQ. HIGHER FREQ. BAND ALARM	NO ALARMS BANDS FOR HIGH FREQ. ENVEL. SPECTRA					
7A, 7H, 7V 8A, 8H, 8V (From Case A)	in/sec	.375	5310	Tapered Roller	300K	800	LOWER FREQ. HIGHER FREQ. BAND ALARM	3000 6400 .340	6400 11,800 .115	11,800 17,000 .095	17,000 65,000 .075	65,000 150,000 .055	150,000 300,000 .040
7H, 8H	gSE	1.20	5310	Tapered Roller	—	N/A	LOWER FREQ. HIGHER FREQ. BAND ALARM	NO ALARMS BANDS FOR HIGH FREQ. ENVEL. SPECTRA					
9A, 9H, 9V 10A, 10H, 10V (From Case G, Type 3 with 15 Vanes)	in/sec	.300	5310	Sleeve	108K	800	LOWER FREQ. HIGHER FREQ. BAND ALARM	1800 4250 .090	4250 9580 .270	9580 20,180 .120	20,180 73,280 .090	73,280 88,000 .210	88,000 108,000 .105

* Positions 6H and 8H must be Stud or Adhesive Mounted for these High Frequency Acceleration Measurements (to 540,000 CPM).

7.16 PERIODIC REEVALUATION OF SPECTRAL ALARM BAND SETUPS ON EACH FAMILY OF MACHINES

Table III gives the procedure on how to specify spectral alarm bands for a number of machine types and bearing configurations. It is intended for use when either originally setting up a PMP database before data is taken, or on the other hand, is intended as a starting point for those who have already taken large numbers of spectral measurements, but have never inserted alarm bands. For those who have specified bands, it serves as a reference to which these setups can be compared.

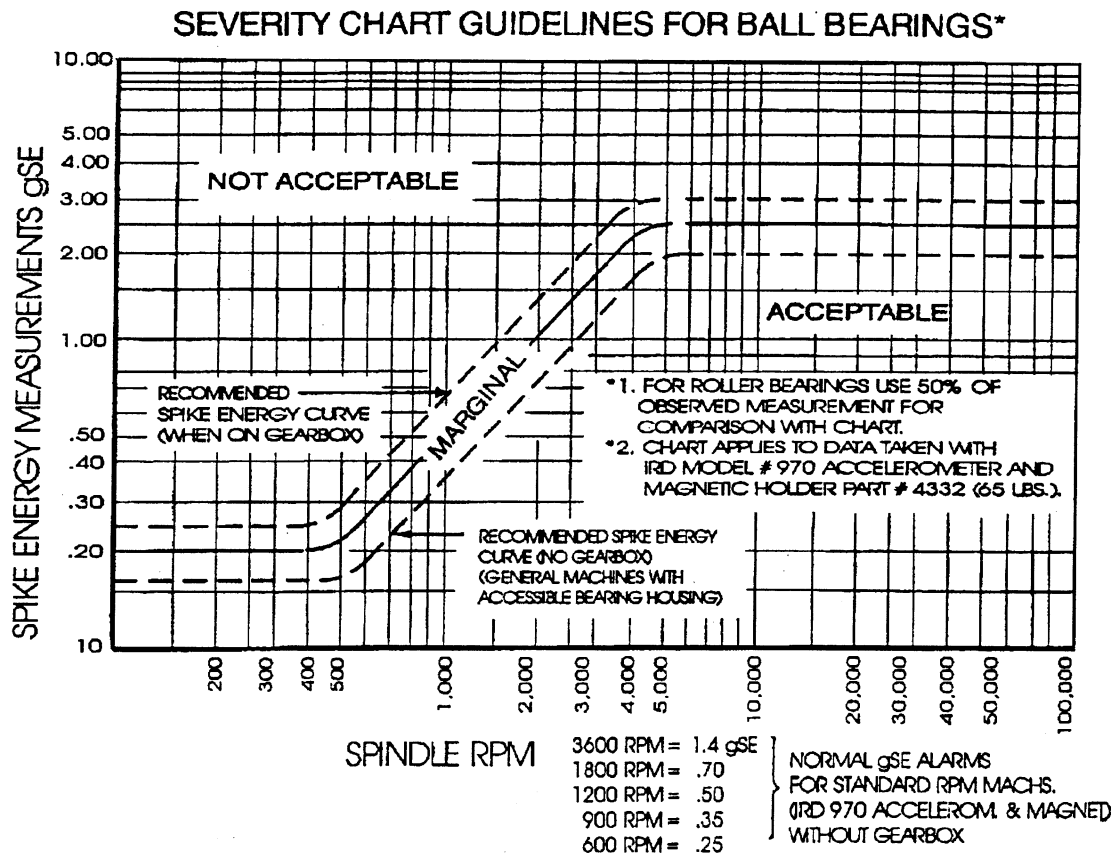
After baselines and perhaps after several additional surveys are completed with these bands in place, the user should reexamine each of the spectral alarm band setups on each family of machines. That is, he should identify certain families, or groupings of similar machines and evaluate how such bands are working, preferably using statistical analysis (24,32, 33,37). He should also evaluate how well the overall alarm levels he has specified for each machine point are functioning. He should ask such questions as:

1. Are the specified alarm bands detecting those problems which are occurring in this machine family - or are they catching some, but missing others?
2. If they are missing some problems, how should they be adjusted in order to better detect them?
3. On the other hand, are these alarm bands causing numerous "false alarms"? If this is allowed to continue, both those in the plant as well as analysts in the program will begin to pay little attention to so-called alarm conditions, even when they actually do detect potentially serious problems.
4. If alarm bands are causing false alarms, how can they be carefully adjusted to noticeably reduce the probability of sounding false alarms, but at the same time, not to the point where they will miss genuine problems?
5. For a given machine family, is one set of spectral alarm bands effectively serving the whole family, or is it necessary to further subdivide the family into two or more groupings? For example, if a utility specified one set of alarm bands for all fans, he will likely find that he needs to further subdivide fans into several individual groups including one set for ID fans, another for FD fans, another for primary air fans, and even another for gas recirculating fans (29).

Finally, it may even be necessary to have one set of alarm bands for the driver and another set for driven pieces of equipment (as was done in Table III, for example, on motor-driven pumps). Or, due to a particular vibration behavior for another family, he may have to specify one set of alarm bands for axial measurements and another for radial (and possibly even different alarm bands altogether for axial, horizontal and vertical measurements).

In general, closely examine each set of spectral alarm bands for each machine family on a quarterly basis (maximum 6 month intervals). Remember, these spectral alarm bands are your "lifeline" with the specific purpose of detecting machine anomalies when they occur, and doing so early in the failure process so that corrective measurements can be orderly planned and so that required replacement parts can be acquired. Properly specified, these spectral alarm bands can and will save one many a headache (not to mention considerable dollar savings as well).

CHART 1 **RECOMMENDED SPIKE ENERGY SEVERITY CHART (IRD SPIKE ENERGY)**



SPIKE ENERGY™ MEASUREMENTS

Energy is generated by repetitive transient mechanical impacts. Such impacts typically occur as a result of surface flaws in rolling-element bearings or gear teeth.

This energy is conducted from its source through various paths to the outer surface of the machine structure, and is seen as a small-amplitude vibration at the surface. Accelerometers coupled to the surface generate corresponding electrical signal.

The accelerometer signals processed by unique filtering and detection circuitry to produce a single "figure of merit" related to the intensity of the original impacts. This figure of merit is expressed in "gSE" units.

SPIKE ENERGY™ gSE readings are measurements which can with experience, be correlated with the severity of the casual surface flaws. Even though gSE readings are affected by the nature of the conductive path between the impact source and the accelerometer, similar machine structures will provide a reasonable basis for comparison between the structures.

The gSE figure of merit has proven to be effective in detecting mechanical defects in meshing gears and rolling element bearings. The gSE measurement, when used in conjunction with conventional measurement of vibration velocity and acceleration, provides early indications of mechanical deterioration.

** When used with magnetic holders, accelerometers must be installed with a light coating of silicone grease and tightened to 40 in-lb. torque.

CHART 2
MAINTENANCE DIAGNOSTICS VIBRATION AND HIGH FREQUENCY
GENERAL TOLERANCE CHART FOR PROCESS MACHINERY WITH
ROLLING ELEMENT BEARINGS^①

Unfiltered Vibration Velocity	Shock Pulse & Acoustic Emission	Microlog HFD	IRD Spike Energy ^②	Severity Level and Alarm Status
Inches/Second Peak	Db	g	gSE	
1.5+	50+	5.0+	3.0+	Danger (Shutdown)
.75 - 1.49	40 - 49	3.0 - 4.99	1.50 - 2.99	Very Rough (Alert)
.40 - .74	30 - 39	1.50 - 2.99	.80 - 1.49	Rough (Alert)
.20 - .39	20 - 29	.75 - 1.49	.40 - .79	Fair (Acceptable)
.10 - .19	10 - 19	.30 - .74	.20 - .39	Good (Acceptable)
.01 - .09	1 - 9	.01 - .29	.01 - .19	Smooth (Acceptable)

① Provided by Charles Berggren of Monsanto

② Spike Energy Amplitudes measured using an IRD 970 accelerometer outfitted with IRD's 2-pole, 65 lb. magnet.

7.161 Procedure For Evaluating the Effectiveness of Specified Overall Alarm Levels and Spectral Bands:

Statistical analysis should be employed on both overall alarm levels and spectral band setups for each family of machines. The advent of the personal computer and statistical software now allows one to perform powerful statistical analysis on large numbers of data points that formerly would have been impractical (24,32,33).

The procedure for evaluating overall alarm levels is summarized as follows:

1. Take each family of machines and work with the latest data captured on each machine in that family (for example, a utility might work with the data taken on each point of the ID fan family).
2. Calculate the average overall level for the family (X_{ave}) by summing the level for each point on each machine in the entire family and dividing by the total number of samples as per the following formula:

$$X_{ave} = \frac{\sum_{i=1}^n X_i}{n} = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n} \quad \text{EQUATION (2)}$$

Where:

X_{ave} = Average or Mean Value

X_i = Value of each individual sample i

n = Number of Samples

3. Calculate the Standard Deviation (S) for the data in this family as per Equation (3):

$$S = \sqrt{\frac{\sum_{i=1}^n (X_i - X_{ave})^2}{n-1}} = \sqrt{\frac{(X_1 - X_{ave})^2 + (X_2 - X_{ave})^2 + \dots + (X_n - X_{ave})^2}{n-1}} \quad \text{EQUATION (3)}$$

Where:

S = Standard Deviation

4. Then, calculate the Statistical Overall Alarm for the family using the following formula:

$$\text{Statistical OA} = X_{ave} + 2S \quad \text{EQUATION (4A)}$$

$$(\text{some prefer OA} = X_{ave} + 3S) \quad \text{EQUATION (4B)}$$

Assuming a normal distribution, approximately 95.5% of the data should fall within 2 standard deviations of the statistical average, whereas 99.7% should fall within 3 standard deviations. As the notes accompanying Equations 4A and 4B state, research and application has shown that a statistical overall equal to only the average level (X_{ave}) plus 2 Standard Deviations should be used during Baselineing and the first few follow-up surveys before the plant has had a chance to take corrective actions (since vibration levels will be higher at that time. Later, after corrective actions have been taken (which should lower vibration levels), Equation 4B should be used adding 3 Standard Deviations to the mean value for the Statistical Overall.

5. Compare the formerly specified overall for this family of machines to the calculated statistical overall alarm. If significantly different, normally choose an alarm level close to the statistical alarm since it is based on actual measurements on your specific family of machines. However, if it is felt that the current statistical alarm was calculated from an extraordinarily high number of machines either in very good condition, or vice versa, in very bad operating condition, then go with an overall alarm between the specified and statistical, but weighted towards the statistical. (Please see the example case history in Table IV which includes several "aberrations" out of the 25 different machine "families" investigated at 4 different plants).

This same procedure can be used for evaluating the effectiveness of each of the spectral alarm bands, whether using 6 bands, or up to 400 in those systems which allow narrowband alarms. In these cases, take one band at a time; calculate the average for the entire family; then calculate the standard deviation for that band; then calculate the statistical level for that band adding either two or three standard deviations to the mean value as desired. Then, compare the calculated statistical level to the alarm level for that band which had formerly been specified, and adjust as desired. Computer software is available for automating this statistical analysis process.

7.162 EXAMPLE - "Statistical Analysis of Overall Vibration Velocity in 4 Client Power Plants Using the Procedure Recommended Above":

Table IV includes results of a statistical analysis performed on overall velocity levels from 25 different machine types ("families") in each of 4 power plants for a client. Data from 442 machines were analyzed as per the statistical procedure outlined above. Not only were the overall levels themselves analyzed, but also the spectral band setups for each of the 25 machine families as well. However, for brevity, only the analysis of overalls is presented here.

Referring to Table IV, note that each of 4 items were tabulated for each plant. First, the quantity of machines in each family was recorded. Next, the overall alarm level used by this plant on each machine family was documented. Then, the average overall for each family was calculated. As few as 5 data collection points were on some machines including motor-mounted fans, whereas as many as 40 data collection points were on some turbine/generators. At this point, no effort was made to separately analyze axial and radial readings, but this step is now being taken. Note that the analysis did separately evaluate turbine/generator gearboxes and exciters from the turbine/generators themselves. Finally, note that the standard deviation was computed for each of the average overall levels.

In general, the study showed that overall alarms used by the client during his baselining were within reason. However, adjustments had to be made to a few.

First, 2 families were found to have excessively high vibration which resulted in high statistical alarms. The 8 turbine/generator gearboxes analyzed had an average .423 in/sec overall and a very high standard deviation of .248 in/sec which resulted in a statistical alarm of .919 in/sec. Similarly, the 7 motor-mounted fans had an average .310 in/sec overall and a high .196 in/sec standard deviation (giving a .701 in/sec statistical alarm using 2 standard deviations). The actual spectrum analysis performed on each of these 2 families of machines found very serious problems in a large proportion of them. Therefore, it was felt that the measured amplitudes were unrealistically high. In the case of the 8 turbine/generator gearboxes, the "recommended alarm" (in the far right-hand column of Table IV) was set at .500 in/sec (up from .440 in/sec). Similarly, the motor-mounted alarm was increased slightly from .325 to .375 in/sec.

On the other hand, several families had much lower vibration than the overall alarms specified by the client. This included each of 7 turbine/generators operating at 1800 RPM; 12 turbine/generators operating at 3600 RPM; 45 vertical condensate/circulating water pumps; 32 forced draft fans; 32 induced draft fans; and 9 general purpose gearboxes. For example, in the case of the turbine/generators, the client used the overall alarms now being proposed by the latest version of ISO 2372 now under consideration which it is understood will call for .300 in/sec peak for 1800 RPM machines and .440 in/sec peak for 3600 RPM turbine/generators.

Here, the average levels were only .097 in/sec for 1800 RPM machines and .147 in/sec for the 3600 RPM turbine/generators. Each of these families had fairly low average standard deviations of overalls which resulted in statistical alarms of .226 in/sec for the 1800 RPM and .339 in/sec for the 3600 RPM turbine/generators (see Table IV). Therefore, recommended alarm levels for both machines were dropped from .300 in/sec to .250 in/sec for the 1800 RPM machines and from .440 in/sec to .350 in/sec on the 3600 RPM machines. These will be further adjusted, if needed, in the future. Similar steps were taken on the vertical condensate/circulating water pumps, forced draft fans, induced draft fans and general purpose gearboxes.

One family obviously had statistically poorly grouped readings with an average overall for 18 general purpose fans being .175 in/sec compared to its standard deviation of .203 in/sec, far above the anticipated norm for a standard deviation compared to this average. Notice that standard deviation levels at both plant 3 and plant 4 were high on these machines. As a result, it was recommended to leave the alarm setting at .375 in/sec until further data has been recorded, and until several machines have been repaired (the spectrum analysis also found a number of problems in these machines which required corrective action).

Finally, please note the bottom row of Table IV entitled "TOTALS". Notice that the average overalls for all machines at each plant were all fairly close to one another ranging from a low of .097 in/sec at Plant 2 up to a high of .128 in/sec at Plant 4. Plant 4 had the largest standard deviation of .120 in/sec. Interestingly, after analyzing data from thousands of points on these 442 machines, the final calculated statistical alarm for all of these machines came out to be a level of

.294 in/sec - very close to the so-called "industry standard" of .300 in/sec which many plants employ as an alarm on all families of machines, regardless of their machine type, speed or drive configuration. However, by examining Table IV, there still was a great disparity between the average overalls of some families.

7.17 CONCLUSIONS

The last few years have brought about dramatic accomplishments and innovative new, powerful tools into the fields of vibration analysis and predictive maintenance. In addition, the rapid development of the personal computer continues to provide the creative team of vibration analyst and computer programmer with an exciting new menu of imaginative ways of not only storing and processing the vast amount of data being gathered, but of effectively putting it to use. This has the effect of multiplying the accomplishments of those within the vibration diagnostics field, and goes far in expanding the horizons of our knowledge. Surely, many more machine "problem sources" will be added to the diagnostic chart of Table I within just the next 5 years, and some of the information now in the chart will be refined. When this occurs, one will be enabled to set up even more effective spectral alarm bands than those provided in Table III. No matter what happens, the effectiveness of every program will depend on the capability of that system to first detect the presence of real problems, and then all available means including software and the knowledge of vibration analysts themselves will be deployed to diagnose both the cause and severity of the problems. Therefore, it now appears that the success of all future systems will depend on how effective the spectral alarm band setups will be in accomplishing this critical problem detection process. It is sincerely hoped that this paper will be of some assistance in helping many to compose exactly this quality of effective spectral alarm bands.

TABLE IV

TABLE IV. STATISTICAL ANALYSIS OF OVERALL VIBRATION VELOCITY IN 4 CLIENT POWER PLANTS

MACHINE TYPE	PLANT 1				PLANT 2				PLANT 3				PLANT 4				CORPORATE SUMMARY (4 PLANTS)					
	QTY.	OA ALARM	AVE.	STD. DEV.	QTY.	OA ALARM	AVE.	STD. DEV.	QTY.	OA ALARM	AVE.	STD. DEV.	QTY.	OA ALARM	AVE.	STD. DEV.	QTY.	OA ALARM	CORP. AVE.		STAT. ALARM	RECOM. ALARM
																			OA	S.D.		
TURBINE GENERATORS A) 1800 RPM B) 3600 RPM	--	--	--	--	2	.300	.078	.041	5	.300	.105	.074	--	--	--	--	7	.300	.097	.065	.226	.250
	4	.440	.161	.093	4	.440	.139	.092	3	.440	.158	.119	1	.440	.084	.055	12	.440	.147	.096	.339	.350
	4	.440	.286	.291	2	.440	.401	.157	2	.440	.718	.253	--	--	--	--	8	.440	.423	.248	.919	.500
	4	.300	.153	.082	6	.250	.044	.049	5	.250	.092	.111	--	--	--	--	15	.263	.089	.078	.246	.250
BOILER FEED PUMPS A) MOTOR B) TURBINE	4	.300	.113	.067	6	.300	.141	.078	9	.375	.155	.092	--	--	--	--	24	.328	.136	.079	.294	.300
	--	--	--	--	2	.300	.148	.079	--	--	--	--	2	.300	.233	.127	4	.300	.191	.103	.397	.375
	--	--	--	--	2	.300	.087	.032	2	.300	.056	.043	2	.300	.144	.049	6	.300	.096	.041	.178	.200
	--	--	--	--	--	--	--	--	--	--	--	--	2	.300	.090	.054	2	.300	.090	.054	.198	.200
BOOSTER PUMPS COND/CIRC WATER PUMPS A) HORIZONTAL B) VERTICAL	8	.300	.040	.015	11	.300	.065	.092	18	.300	.081	.071	8	.300	.069	.047	45	.300	.068	.062	.191	.200
	7	.350	.140	.083	7	.300	.074	.125	7	.300	.064	.051	--	--	--	--	21	.317	.093	.086	.265	.300
	--	--	--	--	--	--	--	--	6	.300	.134	.088	--	--	--	--	6	.300	.134	.088	.310	.300
	12	.300	.128	.092	10	.300	.081	.054	41	.300	.094	.069	21	.300	.127	.123	84	.300	.106	.084	.274	.300
GENERAL PURPOSE PUMPS CENTRIF. AIR COMPRESSORS RECIP. AIR COMPRESSORS	1	.300	.125	.053	--	--	--	--	--	--	--	--	2	.250	.102	.070	3	.267	.110	.064	.238	.250
	5	.425	.149	.054	7	.425	.047	.023	10	.425	.197	.214	4	.425	.125	.075	26	.425	.136	.110	.357	.425
	8	.375	.096	.078	8	.375	.060	.040	12	.375	.063	.054	4	.375	.066	.048	32	.375	.071	.056	.182	.250
	8	.375	.120	.068	8	.375	.045	.024	12	.375	.066	.046	4	.375	.063	.060	32	.375	.074	.048	.169	.250
INDUCED DRAFT FANS MOTOR-MOUNTED FANS PRIMARY AIR FANS ROOTS BLOWERS FLY ASH BLOWERS	--	--	--	--	--	--	--	--	5	.275	.255	.165	2	.375	.446	.272	7	.325	.310	.196	.701	.375
	--	--	--	--	--	--	--	--	11	.375	.132	.159	--	--	--	--	11	.375	.132	.159	.450	.375
	--	--	--	--	--	--	--	--	4	.375	.110	.066	--	--	--	--	4	.375	.110	.066	.242	.300
	--	--	--	--	2	.375	.236	.107	--	--	--	--	--	--	--	--	2	.375	.236	.107	.450	.375
AIR HEATERS GENERAL PURPOSE FANS PULVERIZERS GEN. PURPOSE MACHINES GEN. PURPOSE GEARBOXES	8	.375	.070	.056	--	--	--	--	8	.375	.080	.059	--	--	--	--	16	.375	.075	.058	.190	.250
	--	--	--	--	--	--	--	--	6	.375	.212	.128	12	.375	.156	.241	18	.375	.175	.203	.581	.375
	--	--	--	--	14	.375	.147	.144	26	.375	.128	.094	--	--	--	--	40	.375	.135	.112	.358	.375
	--	--	--	--	--	--	--	--	8	.300	.066	.032	--	--	--	--	8	.300	.066	.032	.130	.200
TOTALS	78	N/A	.121	.080	91	N/A	.097	.078	207	N/A	.116	.098	66	N/A	.128	.120	442	N/A	.115	.090	.294	N/A

NOTES:

- ① = Calculated Statistical Alarms were significantly lower than the Average Alarms used. Thus, Recommended Alarms were reduced to a lower level.
- ② = Calculated Statistical Alarms were significantly higher than the Average Alarms used. Thus, Recommended Alarms were increased to a higher level.

REFERENCES

1. American Gear Manufacturers Association (AGMA); Arlington, Virginia; AGMA Standard 110.04 (August, 1980); "Nomenclature of Gear Tooth Failure Modes"; Pages 6 - 23.
2. Bently Nevada Corporation; Minden, Nevada; Mechanical Engineering Seminar; October, 1984; "Section 6 - Introduction to Rotor Dynamics"; "Section 9 - Rotor Instability"; "Section 11 - Machinery Rubs".
3. Berggren, J. Charles; Monsanto Chemical Company; Pensacola, Florida; "Diagnosing Faults in Rolling Element Bearings, Part I. Assessing Bearing Condition"; Vibrations, Volume 4, No. 1; March, 1988; Pages 5 - 14.
4. Berggren, J. Charles; Monsanto Chemical Company; Pensacola, Florida; "Diagnosing Faults in Rolling Element Bearings, Part II. Alternative Analytical Methods"; Vibrations, Volume 4, No. 2; June, 1988; Pages 12 - 23.
5. Berggren, J. Charles; Monsanto Chemical Company; Pensacola, Florida; "Diagnosing Faults in Rolling Element Bearings, Part III. Electronic Data Collector Applications"; Vibrations, Volume 5, No. 2; June, 1989; Pages 8 - 19.
6. Berry, James E.; Technical Associates of Charlotte, Inc.; Charlotte, North Carolina; "Problem Diagnostics on High-Speed Centrifugal Compressors Using Vibration Signature Analysis"; Proceedings 12th Annual Meeting - The Vibration Institute; Nashville, Tennessee; May, 1988; Pages 1 - 13.
7. Bradbury, E. R.; Union Carbide Industrial Gases, Inc.; Tonawanda, New York; "The Control Chart: A Basic Tool of Statistical Quality Control"; Proceedings 13th Annual Meeting - The Vibration Institute; June, 1989; Pages 87 - 92.
8. Bradley, Dan; IRD Mechanalysis; Columbus, Ohio; "Introduction to FFT Terms and Parameters"; Pages 1 - 9.
9. Bruel & Kjaer; Marlborough, Massachusetts; The Application of Vibration Measurement and Analysis in Machine Maintenance; "The Application of Frequency Analysis to Machine Diagnosis"; Chapter 7, Pages 1 - 12.
10. Bruel & Kjaer; Marlborough, Massachusetts; Piezoelectric Accelerometers and Vibration Preamplifiers Theory and Application Handbook; March, 1978; Pages 50 - 59.
11. Campbell, W. R.; ARAMCO; Dhahran, Saudi Arabia; "Diagnosing Alternating Current Electric Motor Problems"; Vibrations, Volume 1, No. 3; December, 1985; Pages 12 - 15.
12. Corey, Cletus A.; Magnetek, Louis Allis; Milwaukee, Wisconsin; "Induction Motor Electrical Noise and Vibration - Sources and Case Problems"; Proceedings 12th Annual Meeting - The Vibration Institute; May, 1988; Pages 171 - 178.
13. Eshleman, Ronald L.; Vibration Institute; Clarendon Hills, Illinois; "Periodic Monitoring for Predictive Maintenance"; Vibrations, Volume 3, No. 1; June, 1987; Pages 3 - 8.
14. Eshleman, Ronald L.; Vibration Institute; Clarendon Hills, Illinois; "Techniques for the Development of Criteria and Limits for Monitoring Machinery Vibration"; Vibrations, Volume 2, No. 2; September, 1986; Pages 5 - 11.

-
15. Eshleman, Ronald L.; Vibration Institute; Clarendon Hills, Illinois; "Machinery Condition Analysis"; Vibrations, Volume 4, No. 2; June, 1988; Pages 3 - 11.
 16. Hewlett-Packard Company; Palo Alto, California; Dynamic Signal Analyzer Application - Effective Machinery Maintenance Using Vibration Analysis, "Application Note 243-1"; October, 1983; Pages 23 - 44.
 17. Hewlett-Packard Company; Palo Alto, California; The Fundamentals of Signal Analysis, "Application Note 243"; July, 1982; Pages 13 - 17, 25 - 39.
 18. Hydraulic Institute; Cleveland, Ohio; "Acceptable Field Vibration Limits for Vertical Non-Clog Pumps"; Hydraulic Institute Standards for Centrifugal, Rotary & Reciprocating Pumps, 14th Edition, 1983; Figure 78, Page 121.
 19. International Standard Organization; ISO 2372 - "Mechanical Vibration of Machines With Operating Speeds From 10 to 200 rev/sec - Basis For Specifying Evaluation Standards"; First Edition, 1974-11-01; Pages 1 - 7.
 20. IRD Mechanalysis; Columbus, Ohio; Advanced Training Manual, "Vibration Analysis"; Pages 51 - 142.
 21. IRD Mechanalysis; Columbus, Ohio; Vibration Technology - II; 1989; "Systems Dynamics & Resonance".
 22. IRD Mechanalysis; Columbus, Ohio; Vibration Technology - I; 1988; Pages 5-1 thru 6-20.
 23. Jacobs, Ronald W.; Monsanto Company; Addyston, Ohio; "Detection of Mechanical Faults in Rotary Blowers"; Vibrations, Volume 2, No. 1; June, 1986; Pages 9 -13.
 24. Jacobs, Ronald W.; Monsanto Chemical Company; Addyston, Ohio; "SQC And Predictive Maintenance"; Proceedings 13th Annual Meeting - The Vibration Institute; June, 1989; Pages 83 - 86.
 25. Maxwell, J. Howard; Arizona Public Service Company; Palo Verde Nuclear Generation Station; Phoenix, Arizona; "Induction Motor Magnetic Vibration"; Proceedings Machinery Vibration Monitoring and Analysis Meeting - The Vibration Institute, The Vibration Institute, April, 1983, Pages 39 - 51.
 26. Middleton; Ben; Palomar Technology International; Carlsbad, California; "Rolling Element Bearing Failure Detection Methods"; Presented at the Acoustical Society of America, Raleigh, North Carolina, October 8 - 9, 1987; Pages 1 - 14.
 27. Mitchell, John S.; Palomar Technology International; Carlsbad, California; An Introduction To Machinery Analysis and Monitoring; Pennwell Publishing Company; Tulsa, Oklahoma; 1981; Pages 141 - 151, 172 - 204.
 28. Peterson, David; Computational Systems, Inc.; Knoxville, Tennessee; "Vibration Alarm Methods in Predictive Maintenance Programs"; P/PM Technology Volume 3, Issue 1 - January/February, 1990; Pages 22 - 25.
 29. Piety, Kenneth R.; Piety, Richard W.; Computational Systems, Inc.; Knoxville, Tennessee; Scheibel, John R. (Electric Power Research Institute); "Vibration Monitoring of Centrifugal Fans in Fossil-Fired Power Generation"; Vibrations, Volume 6, No. 1; March, 1990; Pages 8 - 13.

-
30. Piotrowski, John D.; General Electric Company; Evendale, Ohio; "Alignment Condition and Its Effect on The Vibration Response of Rotating Machinery"; Vibrations, Volume 1, No. 4; March, 1986; Pages 11 - 17.
 31. Rockland Scientific Corp.; Rockleigh, New Jersey; "Machinery Vibration Diagnostic Guide, Application Note 22".
 32. Smiley, R. G.; Entek Scientific Corporation; Cincinnati, Ohio; "Set Alarm Levels Without Guesswork"; Proceedings 12th Annual Meeting - The Vibration Institute; May, 1988; Pages 107 - 112.
 33. Smiley, R. G. and Schlitz, R. L.; Entek Scientific Corporation; Cincinnati, Ohio; "Statistics or Standards?"; Sound and Vibration; September, 1989; Pages 22 - 23.
 34. Szrom, David B.; Mechanical Consultants, Inc.; Homewood, Illinois; "Analysis and Correction of Gearbox Defects", Proceedings Machinery Vibration Monitoring and Analysis Meeting - The Vibration Institute, June, 1984; Pages 147 - 153.
 35. Szrom, David B.; Mechanical Consultants, Inc.; Homewood, Illinois; "Determining Gear Condition With FFT Spectrum Analysis"; Proceedings 11th Annual Meeting - The Vibration Institute; June, 1987; Pages 1 - 5.
 36. Thomson, W. T. and Chalmers, S. J.; Robert Gordon's Institute of Technology; Aberdeen, Scotland; "An On-Line Computer Based Current Monitoring System for Rotor Fault Diagnosis and 3-Phase Induction Motors"; Pages 1 - 15.
 37. Wetzel, Rick; Entek Scientific Corporation; Cincinnati, Ohio; "Statistical Alarm Methods"; Pages 1 - 4.
 38. Winterton, John G.; Bently Nevada Corporation; Broomall, Pennsylvania; "Component Identification of Gear Generated Spectra"; Proceedings 11th Annual Meeting - The Vibration Institute; June, 1987; Pages 11 - 17.